

EXHIBIT D

**EXPERT REPORT BY JOHN E. HOFMANN
UNITED STATES V. CEMEX, INC.**

**CASE NO. 09-CV-00019-MSK-MEH
IN THE U.S. DISTRICT COURT FOR THE DISTRICT OF COLORADO**

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A handwritten signature in black ink that reads "John E. Hofmann".

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1. EXECUTIVE SUMMARY

Trinity Consultants, Inc. (Trinity) was retained by Crowell & Moring LLP to provide information and opinions for the United States v. CEMEX, Inc. Case No. 09-cv-00019-MSK-MEH.

1.1 INFORMATION REQUIRED BY FEDERAL RULES OF CIVIL PROCEDURE

The following is a listing of the items provided as required by the Federal Rules of Civil Procedure.

- This report contains my opinions and conclusions.
- A statement of my qualifications is contained in Appendix A.
- Appendix B lists the documents, data, or other information considered in forming the opinions contained in this report.
- My billing rates to prepare this report are provided in Appendix C.
- Supporting calculations and exhibits are provided in Appendix D.

1.2 PURPOSE OF REPORT

I was asked by Crowell & Moring LLP (Crowell) to provide opinions on the subjects identified below. I relied on the documents (Appendix B) produced in this case to develop the opinions expressed in this expert report. The Lyons plant is located in Boulder County, Colorado. I have reviewed and provided my opinions on the NO_x emissions and Prevention of Significant Deterioration (PSD) applicability for various projects performed at Lyons Plant. This report contains my opinions on the following topics:

- PSD applicability of various projects performed at the Lyons Plant during the 1997 – 2000 time frame¹;
- Evaluation of overall (“aggregated”) net actual emissions changes during the period of 1997 - 2000;
- Discussion of NO_x variability, NO_x emission factors, and typical PSD approaches at cement plants;
- Evaluation of Best Available Control Technology (BACT) for NO_x applicable during the 1997 – 2000 time frame;
- Lowest Achievable Emission Rate (LAER) applicability for particulate matter less than ten microns (PM₁₀) during the 1997 – 2000 time frame; and
- Economic benefit to CEMEX for not undergoing PSD during this time frame.

¹ I was asked to provide my opinions on projects that occurred in 2000 despite the fact that the U.S. complaint contains projects that occurred between 1997-1999.

I have been provided with access to a vast amount of available information produced in this case to date, including every document cited in the U.S. experts' reports. I made my best possible efforts to consider all information relevant to the subject matters described above. However, should new information come to light as the case proceeds forward, I reserve the right to modify my report accordingly. I have not visited the Lyons Plant.

1.2.1 JOHN E. HOFMANN'S QUALIFICATIONS

In preparing this report, I have relied on my 25 years of experience in the environmental engineering field. I hold a B.S. and M.S. in Chemical Engineering from the Massachusetts Institute of Technology (MIT). I also received an M.S. degree in Atmospheric Sciences from the University of Arizona and have completed executive business training at Dartmouth's Tuck Executive Program. Over my career, I have conducted numerous permit applicability reviews relative to United States Environmental Protection Agency (U.S. EPA)'s New Source Review (NSR) regulations – many for modifications.

With my experience, I have taught numerous professional courses on various air quality regulatory subject matters relative to Federal Clean Air Act (CAA) requirements, PSD, and state equivalent programs. Attendees of my courses have included both the regulated source community and also state and U.S. EPA representatives.

I serve as the President and Chief Executive Officer of Trinity Consultants, Inc., at the firm's head quarters office in Dallas, Texas. As a 25-year veteran of the firm, I have served in several key roles at Trinity, a 300-person environmental consulting and software firm with 30 offices world-wide. The firm is strongly focused on industrial CAA compliance work, with a heavy emphasis involving permitting of new sources including sources at cement plants. Before being selected as President and CEO, I served as the firm's COO.

I have substantial consulting experience and am considered as an authority on CAA regulations, air pollution (source) impact assessments, air pollution control technologies, and environmental management systems. I have also conducted process development work associated with air pollution controls used in oil refining operations and conducted research of the formation of acid rain. My consulting focus is CAA compliance. I have routinely been called upon to formally evaluate prospective capital projects involving upgrades and implementation of new and more efficient process in a variety of ways. I have also created and taught numerous private and customized CAA classes to major industrial companies.

I am a member of the Air and Waste Management Association. Trinity is also a member of Portland Cement Association (PCA). I have participated in the Council of Industrial Boiler Owners (CIBO) Environmental Committee since 1994. In 2010, I was elected to the CIBO Board of Directors. CIBO works closely with the U.S. EPA, the United States Department of Energy, state regulatory authorities, and other governmental bodies to effectuate common-sense environmental regulation.

In my current position, I am called upon to solve the most complex technical and regulatory projects my firm (and clients) may encounter and, especially, those relative to the Portland cement industry and refineries.

1.3 SUMMARY OF OPINIONS

I have arrived at the following opinions based on my review of the circumstances and information regarding the subjects on which I have been asked to provide an opinion:

1. The kiln system at the Lyons Plant at all relevant times was a single stage preheater/precalciner. The kiln was converted from a long dry kiln to the current preheater/precalciner configuration in 1980. The precalciner at the Lyons plant was one of the very first built in the country. Relative to the rotary kiln, precalciner produces very small amounts of NO_x, and under certain configurations (such as the one employed at the Lyons plant), have the capability of destroying NO_x formed in the rotary kiln and reducing overall NO_x emissions from the manufacturing process.
2. Colorado Department of Public Health and Environment (CDPHE) is the responsible air permitting agency for the Lyons plant. As a part of the State Implementation Plan (SIP), U.S EPA approves the Colorado Air Quality Control Commissions Regulations (CAQCC) and CDPHE is considered as a SIP approved agency, responsible for reviewing and issuing PSD permits in the state of Colorado. In 1993, CDPHE issued the kiln and dryer permits and in 1995, CDPHE quarry and crusher permits. The Lyons plant was not subject to any PSD reviews as a part of these permits issuance in 1993/1995.
3. There are many different types of NO_x formed in a cement kiln system, including: thermal NO_x, fuel NO_x, feed NO_x, and prompt NO_x. Thermal NO_x is only formed in the rotary kiln portion of the pyroprocessing system (not the precalciner) and constitutes the vast majority of all NO_x formed in the cement manufacturing process. There are many variables which control thermal NO_x formation in the kiln, but the most important variables are fuel use rates, temperature, system stability, and kiln excess oxygen levels. Because thermal NO_x formation rates are highly variable and heavily dependent on operational parameters within the normal range of control of a cement facility (including temperature and kiln excess oxygen levels, as well as kiln stability), it is extremely difficult if not impossible to attribute any particular increase in NO_x emissions to a specific physical change or change in the method of operation of a cement plant. Moreover, normal variations in kiln maintenance, uptime, raw mix composition, fuel composition, and other parameters play an important role in NO_x formation and can influence NO_x emissions (increases or decreases) absent any physical change or change in the method of operation of a cement plant. Fuel NO_x represents a substantially smaller fraction of the total NO_x produced by a cement kiln and its formation is mostly dependent on the amount of fuel used in the calciner and kiln.
4. Coal is the primary fuel at Lyons plant. Approximately, 50-55% of total coal is combusted at calciner and remaining in the rotary kiln, although this percentage varied over time.
5. It appears that between 2000 and 2007 the proper moisture correction factor was not applied to calculate mass based NO_x emissions. All of a kiln system's NO_x emissions are routed to

- the main baghouse stack and are continuously monitored by a Continuous Emissions Monitoring System (CEMS) to demonstrate compliance with the permit limits. Based on the information reviewed to date, it appears that NO_x concentrations at the Lyons facility post-2000 were monitored as dry parts per million (ppm_{vd}). For a CEMS, based on the moisture correction, monitored wet flow rates are corrected to dry flow rates. Using NO_x and corrected flow, mass based pounds per hour (lb/hr) and tons per year (tpy) are calculated.
6. NO_x emissions from a cement kiln are highly variable. The actual variability of NO_x emissions at the Lyons plant is discussed in this report. Review of NO_x emissions and clinker production levels demonstrate that the two are not linearly related. At Lyons plant, over the years, several improvements were made. These improvements increased the system's efficiency and production capacity with no changes in NO_x emissions. In my opinion, for a plant like Lyons, it is inappropriate to utilize a constant emission factor based solely on clinker production for evaluating the PSD applicability for the projects at issue in this case.
 7. U.S. EPA and many other publications have documented the NO_x variability at cement plants. For a long dry kiln/single stage precalciner kiln, published NO_x emission factors are in the range of 2-12 lb/ton. The Lyons's Plant NO_x emissions are within this range. Historical documents at Lyons plant used 11.94 lb/ton of emission factor to establish the 1993 kiln permit's permitted NO_x emission levels of 2,649 tons per year. Over the years, Lyons plant made several changes that improved (lowered) NO_x emission levels.
 8. The United States experts are incorrect that all of the projects allegedly completed between 1997 and 2000 should have been aggregated for PSD purposes.
 9. The 1997 combustion chamber project should have been analyzed individually for PSD purposes and not aggregated with the Lyons Expansion Project (LEP) because it was an independent project driven solely by an operational problem well known to the plant management years before the other projects were even conceived, planned or budgeted.
 10. In March 1997, the calciner combustion chamber was improved to solve a historical "burnover" problem in which calciner fuel was burned and effectively wasted downstream of the calciner. This project was implemented as an improvement to the calciner combustion and allowed CEMEX to more effectively utilize the thermal output of the fuel injected into the calciner (less wastage and increased thermal efficiency). This would have been expected to improve overall fuel efficiency in the pyroprocessing system, because more clinker could be produced using the same amount of fuel. As discussed in this report, overall fuel based NO_x generation in the calciner most likely decreased as a result of the combustion improvement, or, to the extent it increased, did so by decreasing the amount of fuel combusted in the rotary kiln. In both scenarios, the effect of the change would have been a reduction in total NO_x emissions from the pyroprocess. The rotary kiln's thermal NO_x would

have lowered due to reduced air flow resulting from this project if the fuel split did not change and would have further reduced if the fuel split shifted to the precalciner. Accordingly, this project would not have caused any NO_x increases. Therefore, it was reasonable for CEMEX to assume at that time that no PSD review was required for this project.

11. The 1997 ID fan motor replacement project should not be aggregated with the LEP because it was driven by an emergency situation and was not related to the other projects. There is no evidence that CEMEX included this replacement in its planning process prior to the event. Capital spending decisions made subsequent to the event do not support reliance on the ID fan motor replacement with respect to any of the other alleged projects.
12. The plant's 1,250 hp ID fan motor failed in August 1997 resulting in an emergency shutdown of the kiln. CEMEX management replaced the failed motor with the only readily available motor with a RPM rating matching the existing ID fan. The ID fan and associated fan curve were not changed. At the time of this motor's replacement, it might have met the PSD's Routine Maintenance, Repair, & Replacement (RMRR) exclusion. Due to slightly increased motor HP and associated fan's RPMs, there was a possibility of a disproportionately lower increase in kiln flow and clinker production. Due to vibration and system resistance problems, it is reasonable to assume that CEMEX might not have realized any of this minimal increase which is the conclusion reached in the expert report of Mr. Gerald Young. Regardless, considering a maximum 28 tons per day potential clinker increase and a baseline (August 1995 – July 1997) NO_x emission factor of 7.83 lbs/ton, NO_x emission increases would have been below 40 tons. Therefore, it was reasonable for CEMEX to have assumed at the time that PSD review was not triggered.
13. The Lyons Plant started experimenting with oxygen enrichment (OE) in 1997 when it performed an internal test to measure the effect of oxygen enrichment on various variables, including production rates and NO_x emissions. Their experimental period and process was reasonable.
14. The record contains significant data from the 1997 OE Test. These data are more than adequate to assess the impact of OE on the pyroprocessing system's NO_x generation and facility's overall NO_x emissions with a degree of reliability sufficient for a causation assessment under PSD.
15. Based on CEMEX's 1997 test, oxygen injection into the calciner lowers overall NO_x emissions from the pyroprocessing system. The NO_x pounds per hour emitted from the facility was lower (on an average basis) when oxygen was injected into the calciner, even at significantly higher production rates. Therefore, based on their experimental test data that showed clinker production can be increased with no NO_x emission increases, CEMEX was

reasonable in concluding that it could inject oxygen and increase clinker production without increasing NO_x emissions from the facility.

16. In 1998, after experimenting with the use of oxygen for several months, CEMEX approached the U.S. EPA's PSD's program approved state permitting agency (CDPHE) for an approval to build a permanent oxygen enrichment plant. At CDPHE's request, CEMEX agreed to conduct an additional test to supplement their findings. The duration of the 1998 test was sufficient for CEMEX to supplement previously tested NO_x reductions due to OE injection and provided CEMEX with additional support for its belief that oxygen injection would not increase NO_x emissions or trigger PSD.
17. The air permits issued by CDPHE to the Lyons plant allowed production increases associated with the LEP (other upstream and downstream changes). CDPHE agreed with CEMEX's position that the kiln NO_x emissions would not increase as a result of OE. It was reasonable for CEMEX to rely upon CDPHE's position and not to undergo PSD review. Accordingly, in September 1998, CDPHE issued 2 revised permits (crusher and quarry) and 3 new permits (raw material handling, finish mill system, and CKD handling).
18. The Lyons Plant management reasonably utilized advice provided by its consultant [McVehil-Monnett Associates (MMA)], as well as other available data, to determine that the LEP did not trigger PSD for NO_x. The fact that MMA's preliminary analysis of the LEP calculated a significant net emission increase in NO_x, based solely on the use of emission factors that did not take into account the nature of the projects at issue, but only stated an overly simplistic mathematical calculation with no considerations for NO_x improvements. MMA's subsequently provided more advanced advice to CEMEX, which included a discussion of whether oxygen enrichment resulted in a NO_x increase, further supports the reasonableness of CEMEX's determination that it did not trigger PSD.
19. The raw mill changes did not debottleneck the kiln system. The kiln system at Lyons Plant always remained the bottleneck. Increased production in the raw mill system would not have resulted in any NO_x emissions increases, a conclusion reached by the United States' experts. Even if such changes did debottleneck the kiln system, any production increase was only made possible by oxygen enrichment, which CEMEX had a reasonable basis to believe would not increase NO_x emissions. Thus, CEMEX was reasonable in concluding that these projects did not trigger PSD review.
20. Since the mid 1990s, the Lyons plant has increased cement kiln dust (CKD) recycling (instead of wasting it, the plant utilizes it in the kiln as an improved raw mix). CKD is a calcined material and requires minimal (if any) additional fuel in the pyroprocessing system. Increased recycling of CKD would increase clinker production and reduce NO_x emissions in terms of per ton of clinker. To the extent increased CKD use resulted in higher clinker

production levels and any associated NO_x increases, such increases should be excluded from any PSD analysis because of use of alternate raw material exclusions provided under the PSD regulations. Review of CKD clinker production between 1995 – 2006 time period shows that clinker production levels have remained almost flat. This, along with the other factors affecting NO_x variability, makes any attempt to attribute specific production increases to any given project very difficult. Finally, the recently proposed portland cement Maximum Achievable Control Technology (PC MACT) standards recognize benefits of CKD recycling.

21. During this period, slag and flyash were also tried as alternate raw materials. Use of these alternate materials is part of the raw material mix substitution, which Lyons plant continuously evaluates. PSD regulations exclude use of these alternate raw materials from the definition of modification. Similar to CKD, slag can potentially provide an improvement in NO_x emissions.
22. The addition of a booster fan in 2000 is technically outside the time period alleged in the First Amended Complaint. However, CEMEX could have reasonably concluded that this project did not result in any NO_x increases due to improved thermal efficiency and kiln stability.
23. In the 1997-2000 timeframe, Selective Non Catalytic Reduction (SNCR) was not considered Best Available Control Technology (BACT) for NO_x for portland cement facilities. In 1999, the Colorado Department of Public Health and Environmental (CDPHE) issued a PSD permit for a Greenfield cement plant in Colorado. During that PSD permitting process, SNCR was even not considered as a potential BACT. In 2003, another PSD permit issued by an agency under U.S. EPA Region 8 also did not consider SNCR as a BACT at that time. SNCR was not considered as a BACT until recently.
24. The indirect firing system is typically not considered as an add-on control system for the cement kiln; it is part of the pyroprocessing system's inherent design. In my experience with PSD permitting for cement kilns, indirect firing system itself has not been considered as a PSD BACT. Additionally, the low NO_x calciner/kiln system in the Lyons plant itself would have satisfied the BACT requirements applicable at that time. No additional NO_x controls would have been needed if the facility underwent a PSD BACT analysis at the time of any of the alleged projects.
25. Economic benefit to CEMEX was minimal since BACT at the time was likely good combustion practices and/or inherent precalciner equipped pyroprocessing system and would not have required the installation of additional add-on controls.
26. In my experience with the U.S EPA's nonattainment NSR permitting requirements, the PM₁₀ Lowest Achievable Emission Rate (LAER) requirement would have been triggered only if

the project exhibited a PM₁₀ emissions increase (not including precursors) above the significant emissions threshold for PM₁₀ (15 tons per year).

2. OVERVIEW OF LYONS PLANT AND NO_x EMISSIONS FROM CEMENT KILNS

Portland cement kilns are one of the most complicated chemical reactors made by man. The process converts raw feed (primarily limestone, a source of calcium carbonate) to an intermediate product called “clinker.” Clinker is further processed to make portland cement that is sold for use in final products such as concrete.

The clinker manufacturing process requires extremely high temperatures (gas temperatures >3,100⁰F at the burning zone).² Cement kilns operate continuously and process relatively high volumes of materials. A cement kiln is rotary in nature, which ensures efficient heat transfer and provides for adequate residence times for clinker to be made.

2.1 OVERVIEW OF LYONS PLANT

The Lyons plant manufactures portland cement. Limestone and other raw materials extracted from the quarry are processed through a primary crusher at the Dowe Flats Quarry. Quarried materials as well as other purchased raw materials are stored in storage silos before being processed in the raw mill system. The raw mill mixes and crushes the blended materials and delivers homogenized materials (kiln feed) to storage silos.³

From the homogenization silos, kiln feed is delivered to the calciner portion of the kiln. The calciner is designed for direct firing of coal and/or gas through independent burner pipes on opposing sides of calciner. A form of staged combustion, riser duct firing using coal, is employed at the Lyons Plant to improve operation of the calciner.⁴ Starting in 1997, the Lyons plant started injecting purchased oxygen. Starting in 1999, the Lyons plant start using oxygen from an on-site oxygen generation plant to inject into the swirl chamber directly below the first stage calciner. The calcined material then enters the rotary kiln. The material travels towards the clinker discharge end where additional coal is fired for the clinkering process. The clinker is discharged from the kiln onto the clinker cooler.⁵

The clinker is cooled by large amounts of air that is forced upwards through the clinker bed by undergrate fans. The cooled clinker is then moved to the clinker storage buildings and outdoor piles. The stored clinker is then grinded in the finish mill. Gypsum is also grinded in the finish mill with clinker to produce cement.⁶ In summary, Lyons plant has following operational systems:

1. Dowe Flats Quarry and Crusher System
2. Raw Materials Storage and Mill System
3. Coal Mill System

² Rebecca Battye, Stephanie Walsh, and Judy Lee-Greco, *NO_x Control Technologies for the Cement Industry*, final report; prepared for David Sanders, Ozone Policy and Strategies Group, Office of Air Quality Planning and Standards, EPA; (EC/R Incorporated; September 19, 2000), p. 4.

³ US000003427-538: Operating Permit issued by CDPHE on February 1, 2000.

⁴ Expert Report of Walter Greer, March 1, 2011, page 18.

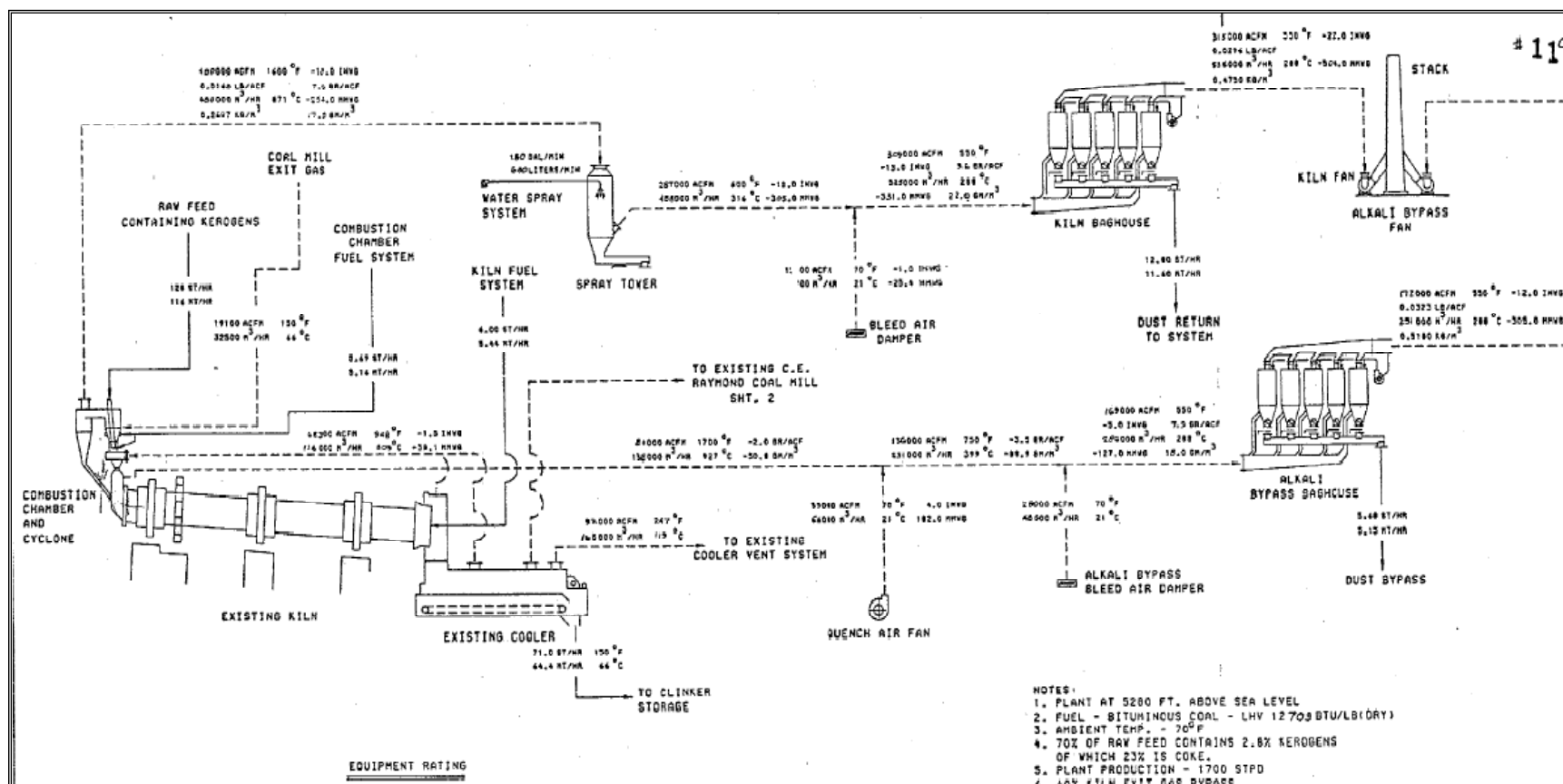
⁵ US000003427-538: Operating Permit issued by CDPHE on February 1, 2000.

⁶ US000003427-538: Operating Permit issued by CDPHE on February 1, 2000.

4. Kiln System: calciner, alkali bypass (ABP), kiln, and associated equipment
5. Clinker Cooler System
6. Finish Mill System
7. Cement Distribution System

Most of NO_x emissions are generated from the pyroprocessing at the kiln system. As shown below in Figure 2-1, all of NO_x generated from the kiln system is vented through a main baghouse. In order to maintain desirable cement quality, some of the kiln system's exhaust gases are also routed to the ABP unit, which has its own baghouse. Therefore, some of the NO_x out of the pyroprocessing system can be vented to the atmosphere via ABP. However, the CEMS, which is upstream to the ABP baghouse monitors all of NO_x emitted from the kiln system.

FIGURE 2-1. KILN SYSTEM⁷



⁷ CEMEX_NEIC_00005738-5753:

2.2 MECHANISMS OF NO_x FORMATION IN CEMENT KILNS

In Portland cement kilns, NO_x is primarily classified into one of two categories: thermal NO_x or fuel NO_x.⁸ In a cement kiln system, multiple raw materials are converted to clinker through a process of simultaneous and sequential chemical reactions. Many variables can affect the equilibrium in the kiln system, which in turn affects the creation of NO_x.⁹

- Thermal NO_x: occurs as a result of the high-temperature oxidation of molecular nitrogen present in the combustion air. Fuel combustion in the rotary kiln itself is the primary source of thermal NO_x formation due to high temperatures above 1400 °C (2550 °F). The fuel combustion temperature in a precalciner is below 1200 °C (2200 °F), which makes thermal NO_x formation impossible. In the precalciner, mostly fuel and feed NO_x are formed rather than thermal NO_x.¹⁰

The gas-phase residence time and the available oxygen concentration are very important parameters in thermal NO_x formation in the kiln itself. Longer residence times at high temperatures will allow the NO to be formed. In addition, greater amounts of oxygen in the combustion zone leads to greater amounts of NO formation.¹¹

- Fuel NO_x: is created by the oxidation of nitrogenous compounds present in the fuel. Coal may contain 1 to 3 percent of nitrogen by weight depending upon the source of coal. The maximum possible fuel NO_x formation may be estimated from the fuel nitrogen content by assuming 100 percent nitrogen conversion.¹²
- Feed NO_x: it is also possible for nitrogenous compounds to be present in the raw material feed and become oxidized to form additional NO_x referred to as feed NO_x. Sometimes, under circumstances where thermal NO_x is relatively low (e.g., in a new modern cement kiln line), the nitrogen in kiln feed can become a noticeable contributor to NO_x formation; however, its contribution is typically minor in comparison to thermal NO_x for long-dry or precalciner cement kilns. The conversion of feed nitrogen to NO_x occurs mainly in the 300 to 800 °C (570 to 1470 °F) temperature range and depends upon the feed heating rate.¹³
- Prompt NO_x: is formed when hydrocarbon (HC) free radicals from the fuel react with available N₂ in the combustion air. The HC free radicals rapidly react with nitrogen to form hydrogen cyanide (HCN) and ammonia. As combustion continues, the HCN and ammonia result in the generation of NO_x. The overall reaction rate of prompt NO_x formation is rapid, and reactions are not strongly temperature dependent and occur in a fuel-rich environment.

⁸ Alternative Control Techniques Document Update - NO_x Emissions from New Cement Kilns, U.S. Environmental Protection Agency, November 2007 (EPA-453/R-07-006), p. 3.

⁹ Walter Greer and Curtis Lesslie, Variability of NO_x Emissions from Precalciner Cement Kiln Systems, 2004.

¹⁰ David Sanders, EPA, NO_x Control Technologies for the Cement Industry (Final Report), September 19, 2000.

¹¹ David Sanders, EPA, NO_x Control Technologies for the Cement Industry (Final Report), September 19, 2000.

¹² David Sanders, EPA, NO_x Control Technologies for the Cement Industry (Final Report), September 19, 2000.

¹³ David Sanders, EPA, NO_x Control Technologies for the Cement Industry (Final Report), September 19, 2000.

The contribution of prompt NO_x to overall NO_x formation in a cement kiln is small because prompt NO_x tends to dominate when the total NO_x concentration in the system is low.¹⁴

Due to the high flame temperature in the burning zone of a rotary long-dry kiln itself (>3,100° F), NO_x emissions from such a kiln are primarily comprised of thermal NO_x. Although NO_x emissions from cement kilns include both nitrogen oxide (NO) and nitrogen dioxide (NO₂), typically, less than 10% of the total NO_x in the flue gas is NO₂.¹⁵ For quantification purposes, the concentration and emission of NO_x are typically expressed in terms of NO₂.¹⁶

Since all of kiln system's exhaust gases are routed to a single baghouse system and monitored by a single CEMS, the amount of fuel NO_x or feed/prompt NO_x formed at different places in kiln is difficult to identify separately from thermal NO_x. Thermal NO_x is assumed to be the dominant mechanism in cement kilns.¹⁷

Long dry kilns typically have only one fuel combustion zone, whereas the precalciner kiln designs have two or three fuel combustion zones: kiln burning zone, riser duct, and precalciner. Because the typical temperatures present in the combustion zones are different, the factors affecting NO_x formation are also somewhat different in different kiln types. The energy efficiency of the cement-making process is an important parameter in determining the amount of heat input needed to produce a unit quantity of clinker. A high thermal efficiency would lead to less consumption of heat and fuel, and would generally produce lower NO_x emissions.¹⁸ Historically, kiln system at the Lyons plant is considered as a thermally inefficient.

A large amount of water in the primary air (from a direct firing coal mill) and injection of cement kiln dust (CKD) in the burning zone (insufflation) may also reduce NO_x per ton of clinker production.

With increasing excess air, the NO_x formation in the kiln will increase, but only up to a certain point as an increasing excess air rate will reduce the secondary air temperature and, consequently, the flame temperature.¹⁹ Per U.S. EPA's ACT document, oxygen levels of 4 to 5 percent in kiln exhaust gases would correspond to high NO_x emissions, whereas oxygen levels of only 0.5 to 1.5 percent would mean lower NO_x emissions. Thus, NO_x emissions in a kiln may depend upon the care exercised by the operator in minimizing excess oxygen needed to maintain the quality of the clinker produced.²⁰

Process conditions that can affect NO_x emissions substantially are temperature stability, stability of raw mix feed rate, burnability of raw mix, and alkali and sulfur control. The excess air used during combustion has a substantial influence on NO_x emissions.

¹⁴ EPA, Alternative Control Techniques Document Update – NO_x Emissions from New Cement Kilns, pg 28

¹⁵ Ibid, p. 6.

¹⁶ David Sanders, EPA, NO_x Control Technologies for the Cement Industry (Final Report), September 19, 2000.

¹⁷ David Sanders, EPA, NO_x Control Technologies for the Cement Industry (Final Report), September 19, 2000.

¹⁸ David Sanders, EPA, NO_x Control Technologies for the Cement Industry (Final Report), September 19, 2000.

¹⁹ David Sanders, EPA, NO_x Control Technologies for the Cement Industry (Final Report), September 19, 2000.

²⁰ David Sanders, EPA, NO_x Control Technologies for the Cement Industry (Final Report), September 19, 2000.

The heating value of the fuel burned may also affect NO_x emissions. Different raw material compositions require different burning conditions to maintain the quality of clinker produced. Thus, similar types of kilns with different feed materials may produce different levels of NO_x emissions.

Attempts to improve energy efficiency of the process by avoiding excessive clinker burning and utilizing waste heat effectively for preheating combustion air, coal, and raw mix is likely to reduce NO_x emissions. The newer preheater and precalciner kiln designs provide very efficient preheating and precalcining of the raw mix with intimate gas solids contact in cyclone towers. The inherent energy efficiency and use of low NO_x calciner of these kiln designs is likely to produce less NO_x emissions per unit amount of clinker as compared to the wet or long dry kilns.²¹

2.3 VARIABILITY OF NO_x EMISSIONS

Kiln operators constantly adjust and account for the changes in the kiln to try and maintain equilibrium in the clinker production process. As a result of these changes, NO_x emissions can vary significantly.²² Factors that affect NO_x formation, such as combustion temperature, fuel nitrogen content and feed rate, raw material nitrogen content and feed rate, and excess oxygen can change rapidly during clinker production. Because these parameters are not in a steady state, corresponding NO_x formation can change rapidly during clinker production.

High variability in NO_x emissions from the kiln system is common to all cement plants.²³ It is for this reason that many kilns have a NO_x emission limit averaged over a month or some longer time period than an hour. The inability of the kiln operator to keep kiln conditions, and thus NO_x emissions, constant results in a difficulty to accurately predict NO_x emissions over short durations and to quantify the short-term NO_x emissions associated with a particular change. Many times multiple changes will occur in a kiln during clinker production. The identification of a single change and its individual impact to overall NO_x emissions is difficult at best. However, it is still possible to undertake short term testing to determine the relative impact of a change on emissions (*e.g.*, the oxygen injections tests conducted at the facility in 1997 and 1998).

The Lyons Plant kiln was originally constructed as a long, dry kiln. The kiln was subsequently modified to a shorter kiln with a preheater/precalciner, and is not directly comparable to other kiln systems.²⁴ Emissions from the Lyons plant's kiln system will most closely resemble emissions from a long dry kiln or an inefficient preheater/precalciner system.

Per Mr. Greer's observations based on his site visit, *"the unusual coal-fired, Lyons pyroprocessing system, with its single preheater has the potential to produce a higher amount of NO_x per ton of clinker than a coal-fired multi-stage preheater-precalciner system of the same clinker capacity.*

²¹ David Sanders, EPA, NO_x Control Technologies for the Cement Industry (Final Report), September 19, 2000.

²² Walter Greer and Curtis Lesslie, Variability of NO_x Emissions from Precalciner Cement Kiln Systems, 2004

²³ Walters, May, Johnson, Macmann, and Woodward, Time-Variability of NO_x Emissions from Portland Cement Kilns, 1999

²⁴ US00031882-31889, pg 1

Primary this is because of the lower thermal efficiency of the Lyons Kiln system and its overall high fuel consumption per ton of clinker.”²⁵

Table 2-1 below provides a summary of NO_x emission rates available for similar types of kilns to the Lyons Plant based on the 1994 Control Technology document compiled by U.S. EPA.²⁶ In January 1995, U.S. EPA published a revised AP-42 section for Portland Cement Manufacturing. The NO_x emission factors adopted for the AP-42 document are also summarized in Table 2-1.²⁷ Each AP-42 emission factor is given a reliability rating from A through E with A being the best. All of the AP-42 NO_x emission factors have a D rating meaning the source tests are based on a generally unacceptable method, but the method may provide an order-of-magnitude value for the source.²⁸

TABLE 2-1. NO_x EMISSION RATES FOR CEMENT KILNS

Process	AP-42 NO_x Factor ¹ (lb/ton)	EPA's Range of NO_x Factors ² (lb/ton)
Wet process kiln	7.4 (D)	3.6-19.5
Long dry process kiln	6.0 (D)	6.1-10.5
Preheater process kiln	4.8 (D)	2.5-11.7
Preheater/precalciner kiln	4.2 (D)	0.9-7.0

¹ U.S. EPA, AP-42, Volume 1, Section 11.6, *Portland Cement Manufacturing*, Fifth Edition, Table 11.6-8, January 1995.

² U.S. EPA, Office of Air Quality Planning and Standards, *Alternative Control Techniques Document Update - NO_x Emissions From New Cement Kilns*, EPA-453/R-07-006, Table 2-1, November 2007.

As Table 2-1 demonstrates, there is substantial variability in the NO_x emissions for kilns. In addition to the operational factors that create variability, there are other factors that create uncertainty when comparing emissions from one kiln to another. These include, but are not limited to, the following:

- Published emission factors from U.S. EPA based on incomplete information;
- Differences in fuel and raw materials over the time; or
- Site-specific testing and kiln types.

NO_x emissions from cement kilns can be calculated using one or a combination of the following methods:

- Using on-site data such as:
 - o Continuous Emission Monitoring System (CEMS)
 - o Compliance Testing/Relative Accuracy Test Audit (RATA);

²⁵ Walter Greer's expert report, March 1, 2011.

²⁶ U.S. EPA, *Alternative Control Techniques Document - NO_x Emissions From Cement Manufacturing*, Table 4-4, March 1994.

²⁷ AP-42 Section 11.6, Table 11.6-8, January 1995.

²⁸ David Sanders, EPA, *NO_x Control Technologies for the Cement Industry (Final Report)*, September 19, 2000.

- EPA's AP-42 Emission Factor (in lb NO_x/ton of clinker produced) – average factors derived from reported emissions;
- Vendor guarantees;
- Data from similar cement kilns; or
- Process knowledge and engineering calculations.

In my experience, NO_x emissions from a cement kiln similar to Lyons would rely on site-specific emission data (stack testing/CEMS). U.S. EPA's lb/ton factors are very generic and do not account for site-specific or project-specific considerations.

Data included in Table 2-2 has been extracted from the May 1994 EPA *Emission Factor Background Documentation for AP-42 Section 11.6* report and September 2000 EPA report, *NO_x Control Strategies for the Cement Industry*, and presents NO_x emission factors for the long-dry cement kiln with Fabric Filter (FF).^{29,30} This data illustrates the variability of NO_x emission rates.

TABLE 2-2. PUBLISHED NO_x EMISSION FACTORS FOR LONG-DRY CEMENT KILN WITH FF

Data Source	No. of Runs	Average NO_x Emission Rate (lb/ton of clinker)	Range of NO_x Emissions (lb/ton of clinker)
AP-42 Data Set 1	2	2.9	1.9-3.8
AP-42 Data Set 2	3	14	14-15
AP-42 Data Set 3	3	9.2	4.5-14
AP-42 Data Set 4	3	4.3	3.4-5.8
AP-42 Data Set 5	6	5.5	4.5-7.3
AP-42 Data Set 6	3	6.7	6.5-6.9
AP-42 Data Set 7	81	6.7	3.4-10
2000 ACT	--	8.6	6.1-10.5

For various reasons, in making assessments of NO_x emission levels for a given kiln, care must be applied to evaluate the various sources of emission data available to calculate an emissions level that is representative of emissions during the baseline emissions period or as post-project potential to emit (PTE) level. Additionally, one must account for inherent variability or project-specific improvements in NO_x emissions, to evaluate any changes in NO_x emissions.

The most simplistic way of evaluating NO_x emissions changes is using clinker production and a fixed lb/ton emission factor without considering project-specific NO_x related reductions or improvements. The problems with this approach include but are not limited to:

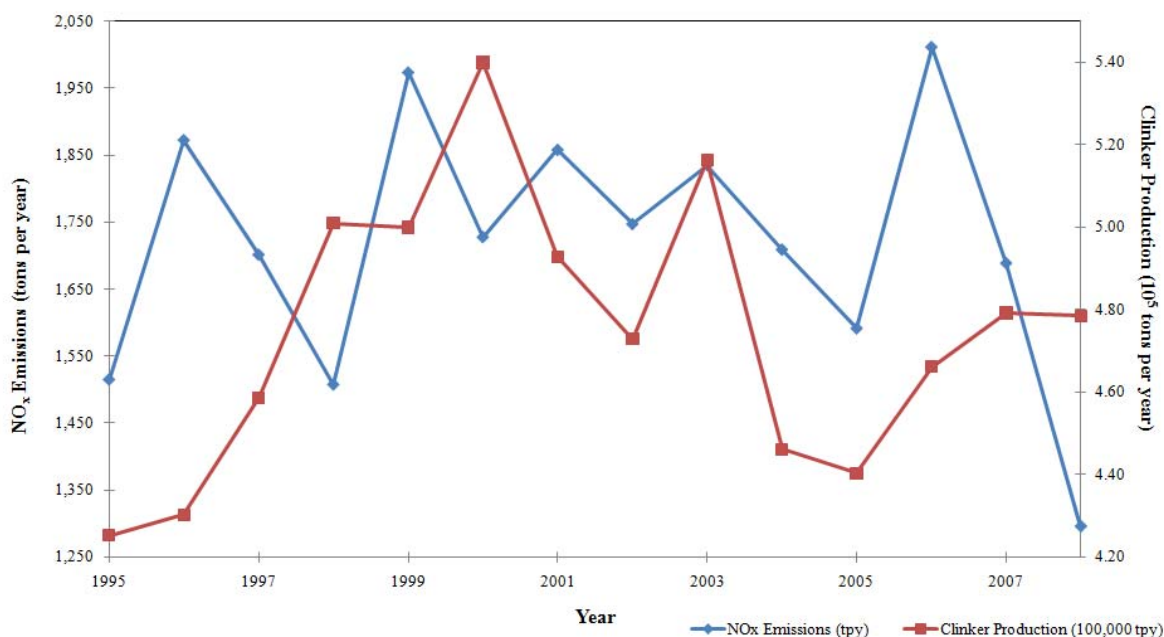
²⁹ EPA, Background Document for AP-42 Chapter 11.6 Portland Cement Manufacturing, May 1994; Table 4-4, pp. 39-40.

³⁰ Rebecca Battye, Stephanie Walsh, and Judy Lee-Greco, *NO_x Control Technologies for the Cement Industry*, final report; prepared for David Sanders, Ozone Policy and Strategies Group, Office of Air Quality Planning and Standards, EPA; (EC/R Incorporated; September 19, 2000), p. 49.

- This approach does not consider any efficiency and improvements type of projects performed at a plant.
- This approach also does not consider inherent variability and factors controlling NO_x emissions, which are not solely dependent on clinker production.

Figure 2-2 below shows clinker production in (100,000 tons) and total corrected NO_x emissions over the years at the Lyons Plant.

FIGURE 2-2. LYONS PLANT'S NO_x EMISSIONS AND CLINKER PRODUCTION VARIABILITY



CEMEX Lyons Facility NO_x Emissions - Kiln - Historical Summary, CMXLYO00628845.

Clinker Production for 1995-2005 is from CEMEX Lyons Facility Estimated Clinker Production - Historical Summary, CMXLYO00628849.

Clinker Production for 2006-2008 is from CMXLYO 52510-3673.

Normalized Clinker Production (10⁵ tpy) = Clinker Production (tpy) / 10⁵

From this chart following observations can be made:

1. NO_x emissions do not have a linear relationship with the clinker production at the Lyons plant. This shows the inherent variability of NO_x emissions, which is dependent on various other operational variables (kiln chemistry, kiln O₂ level, fuel, temperature, etc.). For this reason, it was reasonable for CEMEX to test oxygen injection to determine what if any impact it would have on emissions.
2. Since 1995-2008, NO_x emissions vary from 1,296 tons in 2008 to 2,012 tons in 2006, with an average of 1,717 tons.
3. Over the years, clinker production has a range of ~425,000 (year 1995) to 540,000 tons (year 2000).

4. NO_x is highly variable and is not linearly related to clinker production level. Efficiency improvement projects such as calciner chamber and oxygen enrichment provided additional variability and reductions in NO_x emissions.

Dr. Phyllis Fox in her report mentions that, *“the NO_x emissions from the Lyons Kiln are inherently variable, consistent with results reported for other kilns.” “Curve fitting plus an analysis of variance indicates no statistically significant relationship between the NO_x emission factor in lb/ton and clinker production in ton/yr.”*³¹

We disagree with Dr. Fox’s statement, *“these data indicate that, within the limits of variability and accuracy of measurement, the NO_x emission factor is constant over the range of clinker production.”* As mentioned earlier, the range of the NO_x emission factor (lb/ton) varies from 5.41 to 8.70, which is not at all constant over the range of clinker production. Also, utilizing the NO_x emission factor constant over a long period of time ignores the NO_x reductions and other improvements of a project.

Since NO_x formation in a cement kiln is a very complex process, NO_x emissions calculations solely based on the clinker production levels does not consider the other operational variables.

In order to review PSD applicability for the projects, both U.S. experts utilize an emission factor (lb/ton) based approach to calculate NO_x emission changes without adjusting the NO_x emission factor for the efficiency improvement projects on determining what, if any, emission factor is appropriate. There also has to be a causal relationship between the project and the potential for emissions to be increased. If a source has reasonable basis to believe that a particular physical change or change in the method of operation will not affect emissions, it is reasonable for that source not to undergo PSD analysis.

³¹ Expert report of Dr. Phyllis Fox, March 1, 2011, page no. 18-19.

3. PERMITS AND MOISTURE CORRECTION DISCUSSIONS

This section discusses the permits in place at the Lyons plant during 1997-2000 timeframe as well moisture correction and corrected NO_x emissions.

3.1 PERMITS AT LYONS PLANT

In the early to mid 1990s, the Lyons plant has operated under four separate permits: quarry permit, crusher permit, dryer permit, and kiln permit. In 1998, as a part of the permitting for the so-called Lyons Expansion Project, the quarry and crusher permits were revised. Also, new permits for the finish mill system and CKD handling system were issued in 1998. Table 3-1 below provides a history of these permits and some of the relevant applicable limits.

TABLE 3-1. SUMMARY OF PERMITS DURING 1993 – 1998 TIMEFRAME

Permit Date	Permit No.	Description	Applicable Limits
12/17/1993	12BO444-2	Kiln Permit	- Annual Hours: 8,064 hours/yr - NO _x : 656.9 lb/hr (30-day rolling avg.) - NO _x : 2,649 tpy (12 month rolling avg.)
12/17/1993	12BO444-1	Dryer Permit	- 7,000 hrs/yr
3/15/1995	93BO1414F	Dowe Flats	- Total material: 1,974,507 tpy
3/15/1995	93BO593	Crusher	- Limestone: 800 tons/hr - Limestone: 760,000 tpy
9/11/1998	94-BO-593	Crusher	- Limestone: 800 tons/hr and 400,000 tons/month - Limestone: 1,050,000 tpy
9/11/1998	94-BO-0292	Material Handling	- Crushed Material: 966,000 tpy and 4,000 tpd
9/11/1998	98-BO-0315	CKD Handling	- CKD - 133,000 tpy
9/11/1998	98-BO-0259	Finish Mill	- Clinker - 600,000 tpy and 75,000 tons/month
9/11/1998	98-BO-1414-F	Dowe Flats	- Total Material: 3,200,000 tpy

- As shown in the table, there is no limit on clinker production. There is also no limit on NO_x in terms of lbs/ton of clinker.
- The 1993 Kiln Permit contained a condition that compliance with the annual NO_x limit shall be demonstrated on a rolling 12-month period using the CEMS.³²
- The annual operating hours limit on the kiln system equates to approximately 92.1% uptime (8,064/8,760 hours per year). It appears that in early 1990s, the kiln's uptime performance of was very poor.³³ In 1990, Kiln uptime was approximately 80% and in 1998 approximately 90%. Improvement in kiln uptime likely contributed to clinker production increases at the facility over the years.

³² CEMEX_NEIC_00000203: December 17, 1993 Kiln Permit Conditions 2, 3, and 4.

³³ CMXLYO00150942: CEMEX, Inc. Lyons Cement Plant, Slide No. 4 Kiln Run Time %.

- The 1993 Kiln Permit contained a NO_x limit of 2,649 tpy that was derived from 887,040 tons of an average kiln feed rate.³⁴

All of these construction permits were incorporated into the facility-wide Title V permit issued by CDPHE in 2000.³⁵ The Title V permit acknowledges that the facility used oxygen enrichment and has an onsite oxygen plant.

3.2 CEMS CONFIGURATION

Starting in 1993, CDPHE required the Lyons plant to continuously monitor air pollutant emissions from the kiln baghouse stack, including NO_x. CEMS data are used to determine compliance with applicable tpy emission limits by totaling hourly mass emissions in lb/hr.

The heart of a NO_x CEMS is the analyzer. CEMS installed on the main stack measure concentrations in parts per millions on a dry basis (ppm_{vd}). The CEMS collects data from analyzers into a programmable logic controller (PLC), which is called the Integrated Control Information System (ICIS) data acquisition system (DAS) and will be referred to hereafter as the DAS.^{36,37}

The DAS acquires data, performs calculations, and stores data.³⁸ The DAS at the Lyons Plant was replaced with a WTC, Inc. (WTC) system, which began to be relied upon in January 2000.³⁹ NO_x concentrations are typically determined in the analyzer using the chemiluminescence detection technique. Prior to the analyzer, the gas is dehumidified such that the resulting NO_x concentrations are determined on a moisture free or “dry” basis in ppm_{vd} units. The NO_x mass emission rate can be determined using the measured concentration (provided by the CEMS) and the exhaust gas flow rate.

The flow rate is measured using a separate instrument and moisture is not removed during the flow measurement process. At the Lyons plant, the stack gas flow rate data is measured on a “wet” basis. Since the pollutant concentration is measured on a “dry” basis, the concentration of water vapor (moisture content) in the exhaust gas (in addition to other flow rate parameters) must also be assessed to determine the pollutant mass emissions. Using a dry NO_x concentration with a wet exhaust flow rate will overstate actual NO_x emission by a factor equivalent to the moisture content of the flow rate. Therefore, the DAS at the Lyons plant historically would convert the flow rate from a wet basis to a dry basis in order to calculate NO_x emissions as follows.^{40,41}

³⁴ CEMEX_NEIC_00000203: December 17, 1993 Kiln Permit Condition 3.

³⁵ US000003427-538-c: Operating Permit issued by CDPHE on February 1, 2000.

³⁶ http://www.automatedcontrolsys.com/automatedcontrolsys.com/CEM_Systems_files/CemVocabulary.pdf, date accessed: 4/29/2011

³⁷ http://wtceng.com/cm/?page_id=43, date accessed: 4/29/2011

³⁸ http://wtceng.com/cm/?page_id=43, date accessed: 4/29/2011

³⁹ Sowder’s deposition, pg 89

⁴⁰ Kendrick’s deposition, pg 34

⁴¹ CEMEX_NEIC_00008624: CEMEX CEMS, September 10, 2007.

$$\text{Lbs/hr} = \frac{\text{ppmvd}}{10^6} \cdot \frac{\text{MW lbs}}{\text{lb-mole}} \cdot \frac{\text{lb-mole}}{385.3 \text{ dscf}} \cdot \frac{\text{kwscf}}{\text{min.}} \cdot \frac{10^3}{\text{k}} \cdot [1 - (\% \text{H}_2\text{O} / 100)] \cdot \frac{60 \text{ min.}}{\text{hr}}$$

Where: MW lb/lb-mole = molecular weight of pollutant (CO=28.01, NO_x=46.01, SO₂=64.06)
 385.3 dscf/lb-mole = molar volume of ideal gas at standard conditions
 %H₂O = moisture content of the stack gas, percent wet volume

According to the testimony provided by Monica Sowders, the Lyons plant's WTC system was equipped with automated exhaust gas moisture monitoring.⁴² The software contained in WTC DAS reads the exhaust gas parameters supplied by the CEMS (including and not limited to the dry basis NO_x concentration and the moisture concentration) and the flow rate measurements and calculates the NO_x emission rate. Typically, the pollutant emissions data are archived in an hourly average format.

Monica Sowders testified that CEMEX discovered a flaw in the WTC DAS software installed and used between 2000 - 2007.⁴³ CDPHE's September 2007 letter also confirms that CEMEX over calculated and reported NO_x emissions due to a moisture correction issue.⁴⁴

The WTC DAS software was not "moisture correcting" the NO_x concentrations. Information from WTC confirms that data available to WTC in 2001 indicated that the DAS was not converting the wet flow rate readings to dry readings in order to calculate NO_x emissions.⁴⁵ Accordingly, CEMEX manually corrected the reported NO_x emission rate data back to the date the WTC DAS software was installed. The moisture content used to correct the emission rate data was determined on an annual basis using moisture data collected during the annual Relative Accuracy Test Audit (RATA) stack tests. Moisture concentration measured during the eight RATA stack tests are shown in Table 3-2 below.^{46,47}

TABLE 3-2. MOISTURE CONTENT USED TO CORRECT NO_x EMISSION RATES

Year	RATA Moisture Content^{1,2} (%)
2000	20.1
2001	21.3
2002	19.74
2003	21.4
2004	19.6
2005	18.1
2006	18.3

¹ RATA Moisture Content for 2000 is from CEMEX_NEIC_00008560.

² RATA Moisture Content for 2001 - 2006 is from CMXLYO00131241-131250.

⁴² Deposition of Monica Sowders on January 5, 2011, page 125.

⁴³ Ibid, page 126.

⁴⁴ CEMEX_NEIC_00008624: CEMEX CEMS, September 10, 2007.

⁴⁵ Kendrick's deposition, pg 35

⁴⁶ RATA Moisture Content for 2000 is from "432 - Sowders Exhibit 432".

⁴⁷ RATA Moisture Content for 2001 - 2006 is from "CMXLYO00131241-131250 (Native).xls"

The moisture content varied from 18.1 percent (2005) to over 21.4 percent (2003). An average moisture correction factor of approximately 20% is reasonable based on review of moisture data collected during stack tests at other kiln systems of the same fundamental design as the Lyons plant.

Therefore, it is reasonable for CEMEX to correct the previously reported NO_x emissions produced as a part of this case.

In all of my analyses and discussions, we use CEMEX's moisture corrected NO_x emissions. I did not review actual CEMS data, raw data, DAS logic and correction factor, or RATA, etc. In general, for any CEMS, calculation basis for the flow rate or NO_x concentration needs to be same (dry to dry or wet to wet).

4. PSD REGULATIONS

This section discusses an overview of the PSD regulations effective during the 1997-2000 timeframe, CDPHE's SIP effective at that time, and U.S. EPA's New Source Review (NSR) reform in 2002.

4.1 OVERVIEW OF PSD REGULATIONS

The Clean Air Act Amendments of 1977 enacted the PSD provisions. On August 7, 1980, the U.S. EPA published the version of the PSD regulations that largely formed the backbone of the state rules (SIP rules) implementing PSD during the subsequent two decades. A significant amendment to the 1980 rules occurred in 1988 when an ambient increment limit was added for NO_x (53 FR 40656).

In 1992, the U.S. EPA amended the PSD regulations (57 FR 32314) as a direct result of two court decisions involving the Wisconsin Electric Power Company (*Wisconsin Electric Power Co. v. Reilly*, 893 F.2d 901(7 Cir. 1990) ("WEPCO")) and the Puerto Rican Cement Company (*Puerto Rican Cement Co. v. EPA* 889 F.2d 292 (1st Cir. 1989)). In the WEPCO case, the judge held that the U.S. EPA cannot ignore past operational experience when estimating the impact of a change on future emissions. Specifically, the court rejected U.S. EPA's position on the issue of whether the "actual-to-potential" method should be used to calculate emissions increases for PSD purposes in this case. Consequently, the Seventh Circuit vacated and remanded the PSD determination to U.S. EPA for further action consistent with the court's decision.

CDPHE embodied the PSD regulations into Regulation 3 (5 Colorado Code of Regulations Section 1001-5). The applicable regulations under Regulation 3 were approved by the U.S. EPA. Definitions of the terms "*major modification*", "*net emissions increase*", and "*significant*", contained in the PSD regulations in effect in the State of Colorado during the period 1996 to 2000, are applicable to the assessment of the projects contemplated and completed by CEMEX. These terms are defined in Regulation 3 Part A (I)(B).

There are several important aspects of the PSD regulations that are relevant to an analysis of the projects at issue here. These are:

- 1) The air quality status in the Denver Metropolitan Area (including Lyons) for the period 1996 to 2000,
- 2) The application of the PSD rule exclusions – *e.g.*, specifically the routine maintenance, repair, and replacement exclusion and the fuel/raw material use exclusion,
- 3) The requirement for a causal relationship between a physical or operational change and an emissions increase,
- 4) The criteria used to define a project under the U.S. EPA's aggregation policy,
- 5) The method for determining "baseline" emissions,
- 6) The method for calculating post-change emissions are determined,
- 7) The U.S. EPA's 2002 PSD rule amendments – the reformed PSD rule.

4.1.1 AIR QUALITY DESIGNATION HISTORY – DENVER METROPOLITAN AREA

NO_x

Per first amended complaint filed by the plaintiff, “*at all times relevant to this action, the State of Colorado, including the Boulder County Area was designated as attainment for NO_x, per 40 C.F.R. § 81.306.*”⁴⁸

Therefore, NO_x is considered as an attainment pollutant, subject to PSD review. Accordingly, my report focuses on the alleged projects’ PSD applicability for NO_x emissions.

PM₁₀

Upon U.S. EPA’s promulgation of the PM₁₀ NAAQS in 1987, the Denver metropolitan area was designated as a “Group 1” non-attainment for PM₁₀ due to a history of high Total Suspended Particulate (TSP) readings at area monitors. The area was subsequently designated as a moderate PM₁₀ nonattainment area after the 1990 Clean Air Act Amendments introduced the nonattainment area classification scheme for PM₁₀. The area never exceeded the annual PM₁₀ NAAQS.

In 2001, the CDPHE and the Regional Air Quality Council submitted a PM₁₀ redesignation request to the US EPA. In September, 2002, the US EPA approved the redesignation request and the area was designated PM₁₀ attainment effective October 16, 2002. While the area was designated nonattainment for PM₁₀, the CDPHE identified NO_x and SO₂ as contributors to ambient PM₁₀ concentrations.⁴⁹ As a result, these two pollutants were regulated as PM₁₀ precursors in the Denver metropolitan area.

4.1.2 MODIFICATION EXCLUSIONS - ROUTINE MAINTENANCE, REPAIR, AND REPLACEMENT AND RAW MATERIAL/FUEL USE EXCLUSIONS

The major modification definition contains several exclusions (40 CFR 52.21(2)(iii)). If any of these exclusions apply to a given physical or operational change, the change is not a PSD modification even if there is an associated emissions increase.

The U.S. EPA provides limited formal guidance as to what constitutes “routine maintenance, repair and replacement” (RMRR). From 1990 to early 1999, the U.S. EPA’s loose RMRR guidance was largely framed by the position it took and the arguments it offered in the 1990 case involving the Wisconsin Electric Power Company (WEPCO).⁵⁰ The WEPCO case introduced a five-factor test for determining whether a routine activity could be excluded from PSD permitting.

As such, the U.S. EPA made the case that applicability of the RMRR exclusion was to be evaluated by the source on a case-by-case basis by weighing the nature, extent, purpose, frequency, and cost of the work, as well as other relevant factors to arrive at a common sense finding. U.S. EPA did not expand upon this vague definition until January 1999.

⁴⁸ First amended complaint, filed on March 11, 2009.

⁴⁹ The Code of Colorado Regulations, November 1998 SIP regulations.

⁵⁰ *Wisconsin Electric Power Co. v. Reilly*, 893 F.2d 901(7 Cir. 1990)

The RMRR exclusion does not require that replacement equipment be identical to the equipment replaced. Indeed, replacement equipment will often not be identical to the equipment replaced, but will instead utilize current state-of-the-art technology. For example, a ten year-old piece of equipment does not have to be replaced with an identical ten year-old piece of equipment despite interim technological improvements. If the new equipment serves the same function, there is a reasonable argument that the replacement is routine.

4.1.3 RAW MATERIAL AND FUEL USAGE EXCLUSIONS

The raw material/fuel usage exclusions are embodied in several clauses under the U.S. EPA PSD regulations and Regulation 3 definition of major modification. These are:

1. Use of an alternative fuel by reason of an order or rule under section 125 of the Federal Act;
2. Use of an alternative fuel or raw material by a stationary source which, prior to January 6, 1975 for a source in an attainment or unclassifiable area, and prior to December 21, 1976 for a source in a nonattainment area, the stationary source was capable of accommodating or the source is approved to use under any permit issued under 40 CFR 51 [Subpart I] or 52.21 or under Regulation 3;
3. An increase in the production rate, unless such change would be prohibited under a federally enforceable permit condition;
4. An increase in the hours of operation, such change would be prohibited under a federally enforceable permit condition;

For example, the “increase in the production rate” and the “hours of operation” exclusions have been applied in cases where a manufacturing plant adds a additional operating hours due to economic demand without initiating any physical or operational changes.

4.1.4 CAUSAL EMISSIONS LINKAGE

An important part of the major modification definition is that a physical or operational change must result into an emissions increase above the significance threshold. The Clean Air Act requires only those changes that cause emissions to increase to be to subject to PSD/NSR. If there is no causal relationship between a project and an emission increase, then PSD cannot be implicated. For example, changing the duct work between a plant boiler and the associated stack is clearly a physical change. However, this type of change will have no influence on the quantity of emissions and is therefore not evaluated for PSD permitting applicability.

The U.S. EPA clearly understood this concept when it promulgated the 1980 PSD regulations containing the causation criteria in the major modification definition. The U.S. EPA provided that a “major modification” will occur under the 1980 NSR rules only when there is a **direct causal link** between a non-exempt change and any subsequent emissions increase. In the preamble to the 1992 PSD rule amendments, the U.S. EPA reiterated the causal linkage when it wrote (57 FR 32326):

"The NSR regulatory provisions require that the physical or operational change "result in" an increase in actual emissions in order to consider that change to be a modification (see, e.g., 40 C.F.R. 52.21(b)(2)(i))."

In other words, PSD does not apply unless the source finds that there is a causal link between the proposed change and any post-change increase in emissions.

More recently, the U.S. EPA again confirmed the casual linkage in the December, 2002, version of the PSD regulations. The 2002 regulations articulated the "demand growth" exclusion that originates from the causal link requirement of the Clean Air Act. Under the demand growth principle, only increases that result from a proposed physical or operational method change are to be taken into account (67 FR 80186):

"Both the statute and implementing regulations indicate that there should be a causal link between the proposed change and any post-change increase in emissions..."

Finally, the US EPA again wrote on September 14, 2006, in their proposed rule involving Debottlenecking, Aggregation, and Project Netting (71 FR 54235):

"As we explained in promulgating the demand growth exclusion, we interpret the "which increases" and "which results in" language of section 111(a)(4) of the modification provision of the CAA as requiring "a causal link between the proposed change and any postchange increase in emissions.""

Physical and operational changes are common in a variety of industrial plants. Without the causal relationship required by the Clean Air Act, thousands of projects every year would have to undergo PSD. U.S. EPA itself acknowledges that not every physical change need undergo PSD review and estimates that no more than 200 PSD permits a year are issued. Proving or disproving linkage with emissions can be complicated and often it requires testing when economically and technically feasible. Changes that improve product characteristics (durability, color, hardness, etc.) are often unlinked to emissions. There are many examples where production can be increased without linkage to emissions or result in lower emissions, including.

1. The emergence of continuous casting at steel mills resulted in substantial energy savings and increased production due to more effective utilization of raw materials.
2. Advances in catalyst technology has allowed numerous industries (oil refining, commodity chemicals, etc) to vastly increase product yields with no change in air pollutant emissions.
3. New robotic application devices used in surface coating can cover more area with less coating material due to more precise delivery of material (less "overspray").

Assessing the emissions impact (if any) of new manufacturing technologies, applications of existing technologies to new industries, or simple changes to an existing manufacturing process is a demanding and very frequent challenge for environmental professionals. Therefore, economic and technical realities dictate the use of estimating techniques that are often much more sophisticated than using simple emission factors such as standardized lbs/ton approach, engineering studies, small scale

tests, and experienced based judgment to assess causal emissions linkage and, if so, to estimate the materiality of the emissions increase or decrease.

4.1.5 PROJECT AGGREGATION

Aggregation is the combining of two or more construction projects for the purposes of determining major NSR applicability, and the aggregation rule provides insights on when projects should and should not be combined for NSR applicability purposes. On their face, EPA's PSD regulations do not mandate aggregation; aggregation is only required when a source intentionally splits one project into multiple smaller projects for the express purpose of circumventing PSD/NSR. On June 13, 1989, the U.S. EPA issued a memo stating the agency's position concerning the intentional disaggregation of projects with the intent to circumvent major NSR. When a source owner artificially splits a large project (major modification) into two or more minor projects, the U.S. EPA identifies the resulting minor NSR construction permits as "sham minors." The U.S. EPA's determination that a federally enforceable construction permit is a sham is made based on an evaluation of specific facts and evidence in each individual case. According to the U.S. EPA, the following are criteria that should be scrutinized when making such a determination:⁵¹

- 1. Filing a PSD or nonattainment NSR permit application.*
- 2. Applications for funding.*
- 3. Reports on consumer demand and projected production levels.*
- 4. Statements of authorized representatives of the source regarding plans for operation. ”*⁵²

4.1.6 U.S. EPA's 2009 PROJECT AGGREGATION "RULE"

On January 15, 2009 (74 FR 2376), U.S. EPA published a "final action" on aggregation (rather than a final rule), using a proposal published in 2006 (71 FR 54235) as a starting point. The 2009 action directs sources to combine activities and emissions that are "substantially related" (applying economic and technical dependence considerations) for the purposes of determining NSR applicability. The U.S. EPA previously attempted to define these terms in its 2006 proposed rule, but received comments from a variety of stakeholders stating that the definitions were too prescriptive, potentially leading to increased confusion as compared to the existing policy. U.S. EPA conceded that its attempt to define these terms in a universal manner would be contrary to its acknowledgement that NSR applicability determinations are clearly case-by-case. Thus, the U.S. EPA has not attempted to produce any regulatory language in project-specific terms.

The final action adopted a rebuttable 3-year presumptive timeframe such that two projects separated by more than three years are not substantially related to constitute categorizing them as a single project; however, U.S. EPA views activities separated by less than three years to have no presumption (i.e., not automatically aggregated or disaggregated). In the January 2009 publication, U.S. EPA asserts that there has been some confusion among regulating authorities that closely-timed minor source permit applications are assumed to be aggregated. In contrast, the U.S. EPA describes the

⁵¹ Terrell E. Hunt and John S. Seitz, USEPA, Guidance on Limiting Potential to Emit in New Source Permitting, June 13, 1989.

⁵² See also, "Applicability of New Source Review Circumvention Guidance to 3M-Maplewood, Minnesota" (U.S. EPA, June 17, 1993).

2009 final action as a re-affirmation of its view that operational changes made over a short period of time should be considered for treatment as one project, **but not the sole basis of a decision to aggregate. Thus, U.S. EPA emphasizes that timing alone should not be a basis for aggregation.**

The U.S. EPA's 2009 Project Aggregation rule was slated to be effective on February 17, 2009. However, the new Presidential administration at the time issued a memo withdrawing regulations that were not yet effective. To this date, the U.S. EPA lists the 2009 NSR aggregation rule as "under review."

4.1.7 BASELINE LEVEL OF ACTUAL EMISSIONS

If a project triggers the PSD regulations' "modification" definition then determining the "baseline" level of actual emissions before a physical or operational change is a necessary first step to determine if emissions increase as a result of the physical change. The Agency's regulations define the baseline for PSD purposes, as follows:

"In general, actual emissions as of a particular date shall equal the average rate, in tons-per-year (tpy), at which the unit actually emitted the pollutant during a 2-year period which precedes the particular date and which is representative of normal source operation. The Administrator shall allow the use of a different time period upon a determination that it is more representative of normal source operation. Actual emissions shall be calculated using the unit's actual operating hours, production rates, and types of materials processed, stored, or combusted during the selected time period"

(See 40 CFR 52.21(b)(21)(ii) contained in the PSD rule published on August 7, 1980. The mirror definition is contained in CAQCC Regulation 3, Part A, I.B.1. (1998) and in the CAQCC Common Provisions Regulation I. G. (1995)).

The purpose of the definition is to establish a baseline that is "representative" of "normal" source operations prior to a change. The U.S. EPA historically has followed a presumption that the most recent two years should be used, but has allowed another period where the source demonstrates that recent operations are abnormal (see 45 FR 52676, 52718 (1980)).

4.1.8 POST CHANGE EMISSIONS ESTIMATIONS AND POTENTIAL TO EMIT

Under the US EPA's PSD regulations applicable at times relevant to this case, PSD is triggered if a non-exempt physical or operational change results in a "significant net emissions increase" of a regulated pollutant (40 CFR 52.21(b)(2)(i)). A "net emissions increase" is defined in terms of "any increase in actual emissions from a particular physical change or change in the method of operation at a stationary source" (40 CFR 52.21(b)(3)). "Actual emissions," in turn, is defined as: (1) the average rate, in tons per year, at which the unit actually emitted the pollutant during a two-year period which precedes the particular date and which is representative of normal source operations; (2) the Administrator may presume that the source-specific allowable emissions for a unit are equivalent to the actual emissions of the unit; or (3) for any emissions unit (other than electric utility unit) which has not "begun normal operations" on the particular date, actual emissions shall equal the potential to emit of the unit on that date (40 CFR 52.21(b)(21)). The US EPA takes the position that no unit that

has been physically or operationally changed, regardless of the magnitude, has “begun normal operations” and thus that all modified units must use the actual-to-potential test to determine PSD applicability.

There were inconsistencies in the application of the “actual-to-potential” concept among jurisdictions, the states and U.S. EPA regions during the 1990s. Specifically, complicated scenarios involving physically modified units and un-modified units (which could also experience an actual increase in emissions as a result of the change) were not evaluated consistently among the states and even EPA regions. Partly as a result of this confusion the U.S. EPA embarked upon NSR Reform in 1996 and finalized these regulations in 2002.

4.1.9 NSR REFORM – REVISED PSD REGULATIONS, DECEMBER 31, 2002

On December 31, 2002, the U.S. EPA published revisions to the federal PSD rule commonly referred to as “NSR Reform.” A revised version of the CDPHE Regulation 3 has been adopted by the state of Colorado mirroring the change in the federal rule. A careful comparison of the PSD rule applicable in the late 1990s with the December 31, 2002, rule reveals some of the “changes” under the “reformed regulations” were already allowed, in some form, under the former regulations. Flexibility under the new PSD rule in the areas of baseline emissions determinations and net emissions increase calculation methodology are paralleled by similar provisions under the old PSD regulations.

After determining whether a physical change or change in the method of operation will take place, the foundation for any PSD applicability analysis is the baseline actual emissions determination. Historically, the immediately preceding two years actual emissions were utilized as the baseline emissions. However, as discussed above, the regulations stated the Administrator had the discretion to allow a different time period if it were more representative of normal source operations. In fact, the reformed rule is more prescriptive in that it specifies a consecutive 24-month period instead of presumably allowing the Administrator accept a different time frame than 24 months, as is implied under the definition of “actual emissions” (40 CFR 42.21(b)(21)(ii)).

Defining “net emissions increase,” is a critical element of any PSD applicability analysis. “Net emissions increase,” as traditionally defined in 40 CFR 52.21(b)(3) and the mirror provision in the historical CDPHE Regulation 3, is determined by assessing the increase in ACTUAL emissions from a particular change. This means that if an appropriate framework is set for determining “future” actual emissions then a true “actual-to-actual,” rather than the policy-based “actual-to-potential,” calculation could have been performed in determining the need for PSD permitting. The reformed PSD regulations clarified that this framework applies to non-utility and utility units alike.

4.2 NET EMISSION INCREASES

If a project triggers the modification definition under PSD regulations, U.S. EPA specifically recommends following a six-step procedure to determine the net emissions changes associated with the project. Table 4-1 provides a summary of this six-step procedure.⁵³

**TABLE 4-1. NEW SOURCE REVIEW WORKSHOP MANUAL,
PROCEDURES FOR DETERMINING THE NET EMISSIONS CHANGE AT A SOURCE**

Step 1. Determine the emissions increases (but not any decreases) from the proposed project. If increases are significant, proceed; if not, the sources are not subject to review.
Step 2. Determine the beginning and ending dates of the contemporaneous period as it relates to the proposed modification.
Step 3. Determine which emissions units at the source experienced (or will experience, including any proposed decreases resulting from the proposed project) a creditable increase or decrease in emissions during the contemporaneous period.
Step 4. Determine which emissions changes are creditable.
Step 5. Determine, on a pollutant-by-pollutant basis, the amount of each contemporaneous and creditable emissions increase and decrease.
Step 6. Sum all contemporaneous and creditable increases and decreases with the increase from the proposed modification to determine if a significant net emissions increase will occur.

⁵³ New Source Review Workshop Manual, Prevention of Significant Deterioration and Nonattainment Area Permitting (DRAFT), United States Environmental Protection Agency, Office of Air Quality Planning and Standards, October 1990, pages A.45 to A-51.

5. EVALUATION OF PROJECTS DURING 1997-2000 TIMEFRAME

I reviewed PSD applicability of various projects allegedly undertaken at the Lyons Plant between 1997 – 2000 as identified by Mr. Greer and Dr. Fox.⁵⁴ Both U.S. experts consider all of the projects that occurred at the plant during this period as a part of the “Lyons Expansion Project.” As discussed in this section, my opinion is that not all of these projects should be considered part of the Lyons Expansion Project and would not have been properly aggregated for PSD purposes.

U.S.’s first amended complaint alleges that, “53. *over the course of its operation, the Facility has undergone numerous changes and upgrades. This action focuses on a modification Defendant undertook at its Facility between 1997 and 1999, which resulted in violations of the CAA and the applicable implementation plan, and on its operations subsequent to the completion of this modification.*” Therefore, based on this, the focus of my review was the projects that were initiated during 1997-1999 timeframe. We did not extensively review the changes that occurred after 1999.

In summary, in my PSD applicability evaluation, it is my opinion that the following projects should be evaluated independently and should not be aggregated for PSD purposes:

1. March 1997: Calciner Chamber Improvement Project
2. August 1997: ID Fan Motor Replacement Project
3. 2000 Annulus Boost Fan Project

I believe that the following projects may need to be aggregated for PSD purposes:

1. 1997 – 1999: Purchased Oxygen Enrichment (OE) – considered as a kiln system change
2. 1999 – 2000: On-site OE plant – considered as a kiln system change
3. 1997 – 1999: Raw mill system changes – transfer line, tempering tee damper, and compressor - considered as upstream system change
4. 1998 – 1999: Finish mill system changes – separator - considered as downstream system change
5. 1998 – 1999: Clinker Cooler system change – grates – considered as a downstream system change
6. 1998 – 1999: Dowe flats quarry expansion project - considered as upstream system change

5.1 CALCINER CHAMBER IMPROVEMENT PROJECT

In the 1980s, the original old long dry kiln was converted into one-stage preheater/precalciner kiln. This modification was mainly done to overcome blue haze problems. However, after this conversion, the Lyons plant started observing poor fuel efficiency and an instable calciner combustion system resulting in an over burning of fuel into the cyclone.⁵⁵ The 1996 Fuel Combustion & Technology (FCT) study indicates, “*since the conversion, the plant has always had problems with the cyclone*

⁵⁴ Walter Greer and Dr. Phyllis Fox’s expert reports, dated March 1, 2011.

⁵⁵ CMXLYO00402635: Case Study of Physical Modeling at Southdown’s Lyons Plant, February 12, 1999.

*operating at higher temperatures than the calciner vessel, this phenomena is defined by the plant as turnover.*⁵⁶

In the Lyons pyroprocess, raw feed material is pre-calcined in the calciner before being further calcined in the kiln. This calciner was one of the first calciners in the country and it was very inefficient, requiring almost 5.2 million BTU to produce one ton of clinker.⁵⁷ The primary fuel used at the Lyons plant is coal. The fuel distribution between the calciner and kiln before the implementation of the calciner chamber modification project was approximately 50/50%.⁵⁸ Mr. Greer points out, *“at the Lyons Plant, the fuel split usually is about 50-50% between the kiln and calciner, but it varies as much as 5%, depending on process conditions. The degree of limestone calcinations achieved in the plant’s calciner is in the range of 70-75%.”*⁵⁹

Mr. Lohr’s described the turnover situation during his deposition as follows, *“a lot of fuel was wasted. A lot of that fuel did not burn here, where it would have helped to produce clinker, with calciner material help us make clinker. It would burn in the swirl chamber adjacent or into the spray tower, which is the next step down the process.”*⁶⁰ This problem resulted in inefficient fuel use and fuel being combusted elsewhere, thus not being used effectively.⁶¹

In 1996, FCT conducted modeling of the combustion chamber and from the results of this modeling provided recommendations on refractory improvement to allow thorough mixing of fuel and combustion air in the vessel. Per FCT modeling report, *“modifying the internal profile above the main burners in the calciner vessel can improve the fuel/air mixing.”*⁶² However, this modification would have also increased the static pressure and reduce the air flow through the calciner.⁶³ Per Mr. Young’s expert report mentions, *“Based on these two precautionary statements, FCT was likely aware that their suggested efficiency improvement would increase the static pressure and reduce the air flow rate through the calciner.”*⁶⁴

Mr. Young in his expert report also mentions that:

*“This efficiency improvement permitted the Lyons plant to burn fuel more effectively in the calciner without the “turnover” problem (and without the corresponding high CO emissions). Because the fuel was now being burned in the appropriate place in the system, CEMEX could process more feed and produce more clinker with the same quantity of fuel. However, the total amount of fuel that could be burned in the overall pyroprocessing system would have been reduced because of the reduction in air flow due to the static pressure increase across the calciner.”*⁶⁵

⁵⁶ Physical Model Study of Precalciner, FCT, August 2, 1996.

⁵⁷ CMXLYO00425024: Preliminary Analysis by Colorado APCD on February 21, 1988 for Tire Fuel Permitting.

⁵⁸ CMXLYO00453794: FCT Modeling Report, August 2, 1996.

⁵⁹ Walter Greer expert report, March 1, 2011, page 25.

⁶⁰ Deposition of John Lohr on October 11, 2010, pages 48-56.

⁶¹ Physical Model Study of Precalciner, FCT, August 2, 1996.

⁶² Physical Model Study of Precalciner, FCT, August 2, 1996.

⁶³ CEMEX’s expert report of Mr. Gerald Young.

⁶⁴ CEMEX’s expert report of Mr. Gerald Young.

⁶⁵ CEMEX’s expert report of Mr. Gerald Young.

Based on these recommendations and to resolve burnover issues, the calciner chamber's refractory wall was improved in Spring 1997 by adding refractory.⁶⁶ Per Mr. Lohr's deposition, BTU requirements at Calciner significantly went down, *"we went down to 4.9, 5.0 from that 5.2, approximately 5.1, 5.3 range. Yes we saw some reduction in the millions of BTU required per ton. We did not see a fuel increase. We saw an efficiency increase. The fuel efficiency improved. Fewer tons per ton, fewer MMBtus per ton, so I could make more clinker with the same fuel."*⁶⁷ Additionally, review of the historical MMBtu/ton levels indicate that the 1997 MMBtu/ton thermal efficiency (4.81) went down from 1996 level (5.24) and 1995 (5.15), showing an fuel efficiency improvement of approximately 8%⁶⁸

Fixing the burnover issue would have enabled CEMEX to use coal in the calciner that previously was being burnt at the cyclone or other downstream structures. This would have increased the effective utilization of the coal's MMBtus to produce the additional amount of clinker; therefore, improving overall fuel efficiency of the pyroprocessing system with the possibility of additional clinker production. Increase in fuel efficiency would have resulted in an overall decrease in the NO_x emissions.

As discussed earlier in the NO_x variability section; formation of fuel NO_x in the kiln and the calciner and thermal NO_x in kiln is mainly dependent on the fuel usage. This project would have also improved the calcination process (more of calcination process being performed in the calciner vs the kiln). Therefore, it is reasonable to assume that after this project, fuel split between kiln and calciner might have improved (more in calciner vs. kiln). Mr. Goodrich is his testimony points out that the current fuel split between calciner and kiln is 55-45%.⁶⁹ Per Mr. Young's expert report, more fuel usage in calciner versus the kiln will reduce overall NO_x emissions.

5.1.1 PSD APPLICABILITY OF CALCINER CHAMBER IMPROVEMENT PROJECT

Project Aggregation:

This project was initiated as a result of historical burnover and inefficient combustion issues at the calciner. This project was not part of any clinker production increase efforts later planned at the Lyons plant. This is consistent with the CEMEX's depositions. Mr. Goodrich testified that, *"they [LEP and Calciner Chamber] were entirely independent projects."*⁷⁰ Applying economic and technical dependence considerations, this project is separated from the Lyons Expansion Project. It was also not part of any of the documents prepared as the Lyons Expansion Project. Therefore, in my opinion, this project should not be aggregated with any other projects listed by the U.S DOJ experts or complaints filed by the U.S.

⁶⁶ Deposition of John Lohr on October 11, 201, Exhibit No. 203.

⁶⁷ Deposition of John Lohr on October 11, 2010, pages 48-56.

⁶⁸ CEMEX_NEIC_00008493_Lyons Kiln History and Forecast.

⁶⁹ Deposition of Steve Goodrich on December 9, 2010, pg. 42.

⁷⁰ Deposition of Steve Goodrich on December 9, 2010, Page N0. 53.

NO_x Emissions Changes:

The FCT study performed in 1996 mentions that the efficiency would have improved after this project. As well, depositions by John Lohr, Randy Wiley, and Steve Goodrich also discuss the efficiency improvement. Improved fuel efficiency should have resulted in an overall NO_x decreases.⁷¹ Since it is reasonable to assume that this project would have reduced overall NO_x; no PSD evaluation would have been as needed at that time in 1997.

The U.S EPA and Colorado's PSD regulations provide:

*"The NSR regulatory provisions require that the physical or operational change **"result in"** an increase in actual emissions in order to consider that change to be a modification (see, e.g., 40 C.F.R. 52.21(b)(2)(i))."*

In other words, as discussed earlier in Section 4.1.4, PSD does not apply unless the source finds that there is a causal link between the proposed change and any post-change increase in emissions.

FCT's initial 1996 report did not quantify any clinker production increase due to this project. As discussed earlier due to static pressure issue, this project would not have caused any clinker production increase.⁷² However, review of several other documents, which were internally compiled by various CEMEX employees after the implementation of this 1997 project, indicates a range of 5.8-10% potential clinker increase attributable to this project.^{73,74} What is important, however, is what CEMEX and FCT believed and expected at the time of the project; what someone may have speculated after the fact. Additionally, due to improved fuel efficiency, overall NO_x emissions would have gone down even with this potential clinker production increase.⁷⁵ Thermal NO_x and fuel NO_x formation in the pyroprocessing system is dependent on the fuel usage. Considering static pressure issue and NO_x improvements, it would have been reasonable at the time that NO_x would not have significantly increased; therefore no PSD evaluations would have been needed at the time.

The U.S. EPA's NO_x control technology document predicts a worst-case fuel NO_x emission factor of 1.5 lbs of total fuel NO_x/ton of clinker.⁷⁶ Considering 50-50% split between kiln and precalciner fuel at the Lyons plant before implementation of this project, the fuel NO_x factor will be equal to 0.75 lb of fuel NO_x at calciner or kiln/ton of clinker. Even ignoring any fuel efficiency improvements, if one would have considered worst-case 10% potential clinker production increase from this project, fuel NO_x increase from the calciner alone would have been well below 40 tons of PSD significance threshold (16.48 tons of NO_x = 43,949 tons of clinker increase*0.75 lb of NO_x/ton of clinker/2,000).⁷⁷ Therefore, it is reasonable to assume that no PSD evaluations were needed at that time. Even if one

⁷¹ Expert report of Gerald Young, May 10, 2011.

⁷² Physical Model Study of Precalciner, FCT, August 2, 1996.

⁷³ Deposition of John Lohr on October 11, 201, Exhibit No. 201.

⁷⁴ CMXLYO00402635: Case Study of Physical Modeling at Southdown's Lyons Plant, February 12, 1999.

⁷⁵ Expert Report of Gerald Young, May 10, 2011.

⁷⁶ NO_x Control Technologies for the Cement Industry, by Rebecca Battye, September 19, 2000.

⁷⁷ CEMEX_NEIC_00008526: Lyons Kiln Production and Oxygen Use, January 4, 1998, 54.5 tph historic kiln's production rate and CEMEX_NEIC_00000203 Kiln Permit's 8,064 operating hour limit results in a maximum production rate of 439,488 tons per year (54.5 tph*8,064 hours/yr). 10% of this value equates to 43,949 tons.

considers that all fuel NO_x (kiln and calciner) would have changed as a part of this project, net increases using 1.5 lb/ton overall fuel NO_x factor, would still have been well below 40 tons. It is also reasonable to assume that either thermal NO_x in the kiln would have remained unchanged or decreased.

Actual NO_x Emissions Before and After the Project

Table below shows annual average NO_x emissions using 24-months of before (March 1995 – February 1997) and after (March 1997 – February 1999) this project's implementation in March 1997.

TABLE 5-1. NO_x EMISSIONS 24-MONTHS PRIOR AND AFTER THE CALCINER IMPROVEMENTS

Year		Total NO_x Emissions¹ (tons)	Clinker Production² (tons)
03/1995 - 12/1995	Before the Project	1,268.1	357,600
01/1996 - 12/1996		1,872.5	430,272
01/1997 - 02/1997		360.5	73,573
Annual Average³		1,750.6	430,723
03/1997 - 12/1997	After the Project	1,340.9	385,121
01/1998 - 12/1998		1,507.3	501,008
01/1999 - 02/1999		270.9	65,651
Annual Average³		1,559.6	475,890
Project Increase⁴		-191.0	45,168

¹ CEMEX Lyons Facility NO_x Emissions - Kiln - Historical Summary, CMXLYO00628845.

² CEMEX Lyons Facility Estimated Clinker Production - Historical Summary, CMXLYO00628849.

³ The Annual Average is the sum of 24 month time period divided by 2.

⁴ The Project Increase is the pre-project and post-project annual averages.

As shown in the table, actual NO_x emissions before and after this project show that project was able to achieve intended efficiency improvements and overall NO_x emissions went down after the implementation of this project, which is consistent with what CEMEX believed at that time that this project would not result in any significant NO_x emission increases. Therefore, considering the actual NO_x emissions data available after the implementation of this project, there appears to be no PSD issues associated with this project.

It is my opinion that it was reasonable for CEMEX to have concluded that the combustion chamber project could not causally increase NO_x significantly and thus no PSD review was required. It is further my opinion that had a formal PSD review been undertaken, it would have shown that the project itself would not have increased emissions and hence PSD was not triggered.

5.2 ID FAN MOTOR EMERGENCY REPLACEMENT

As a part of the conversion of the long old dry kiln to an one-stage preheater/precalciner kiln in 1980s, a 1,250 hp motor on the Kiln ID fan was installed.⁷⁸ Typically, operating life of this type of fan motor is 10-20 years depending on alignment, quality of bearings, and amount of vibration to which it is subjected.⁷⁹ From review of various documents, it appears that the fan motor was “amped out” and had vibration issues.⁸⁰ On August 8, 1997, the motor burned up and created an emergency situation at the plant.⁸² This motor was near the end of its operating life.⁸³

Immediately following this motor failure, CEMEX began a diligent search for an identical replacement matching voltage and rotations per minute (RPM) requirements.⁸⁴ After extensive search, CEMEX was only able to locate a single motor replacement with an identical RPM rating. Per Mr. Lohr’s disposition, “*Same RPMs, Motors, AC motors come in typical RPM steps: 1,800, 3,600, what have you.....but it was the same motor. That was one of things that we had to match up*”⁸⁵. Prior to the replacement, the average speed setting for the fan was approximately 1,030 RPM.⁸⁶

The situation at the time of this motor’s replacement was an emergency. The fan motor failed suddenly and CEMEX immediately replaced it with an identical motor. Per Mr. Lohr, “*we looked for a 1250 hp motor or explored with Reliance, who was our motor service vendor at that point in time, how quickly to rebuild what we have, where can we have a replacement motor, what have you. The only motor available in the entire country was a used 1500 hp Louis Allis motor, and so we chose to do that versus waiting a week to ten days or longer to get the existing motor rewound. Motor was damaged very badly.*”⁸⁷ Based on this testimony, it appears that CEMEX looked for an identical replacement available at the time and that the only match that CEMEX was able to find was a motor with 1,500 hp with similar RPM rating range by the same manufacture Louis Allison.⁸⁷

After installation of this motor, CEMEX realized that the fan motor could have provided slightly more RPMs than the original motor. The ID fan induces draft air, which is essential to the flow in the kiln system and clinker production. Based on the review of RATAs performed after the replacement, average RPMs of 1,070s during the RATAs were slightly higher than previous average of 1,030

⁷⁸ Deposition of John Lohr on October 11, 2010, page no. 62.

⁷⁹ Deposition of John Lohr on October 11, 2010, page no. 62.

⁸⁰ Deposition of John Lohr on October 11, 2010, pages 60-80.

⁸¹ Deposition of Randy Wiley on November 10, 2010, page 160.

⁸² CMXLYO00453897:John Lohr’s Memo to the Emergency Breakdown Team on August 11, 1997.

⁸³ Deposition of Steve Goodrich on December 9, 2010, Page No. 77.

⁸⁴ Deposition of John Lohr on October 11, 2010, pages 60-80.

⁸⁵ Deposition of John Lohr on October 11, 2010, pages 60-80.

⁸⁶ CEMEX_NEIC_00008722-9039 and CEMEX_NEIC_00009049-9244: 1995 and 1996 RATAs.

⁸⁷ Deposition of Steve Goodrich on December 9, 2010, Page No. 77.

RPM.⁸⁸ The design specification listed a fan speed of 1,150 RPM.⁸⁹ Based on the review of several documents produced in this case for the ID fan motor replacement, I made the following observations:

- The old Id fan motor was initially installed in 1984 and was close to the end of its operating life.
- The motor suddenly failed on August 8, 1997 and created an emergency situation because of heavy production demands at the time.
- CEMEX made significant efforts given the situation for finding an identical replacement and the closest match to the failed motor was a 1,500 hp motor in the same RPM rating range by the same manufacturer.
- This replacement did not change the fan or associated fan curve, which influences kiln flow.⁹⁰
- Considering the situation surrounding this replacement and due to the fact that the fan and fan curve were not changed, at the time this motor was replaced, it would have been reasonable for CEMEX to assume that no extra RPMs/will be gained from the replacement.⁹¹

Impact of Higher Motor Capacity:

After the replacement in 1997, CEMEX operators may have had the availability of minimal extra RPMs (~40 RPM or 3.9% increase); however, these extra RPMs would not have substantially increased any clinker production level because of other resistance in the kiln system.⁹² Per Mr. Young's expert report, *"the entire 3.88% additional gas flow rate through the fan does not mean that 3.88% additional fuel could have been burned with a corresponding 3.88% increase in clinker production."*⁹³

Mr. Goodrich mentions, *"as rpms are increased on the fan wheel, system resistance also increases, and there is actually an effective null point where it does not matter much how much faster that fan wheel spins, it can not because, it can not overcome that resistance and pull any more air through the system."* *"As I said, to the extent there was any, we are uncertain whether there was an increase. If there was, it was likely minimal"*⁹⁴.

It is also clear that the plant manager at the time of the change, John Lohr, did not believe that the change was being undertaken to increase capacity. As he said in his deposition, he was replacing the fan motor only because the old motor burned out and he replaced it with the only comparable motor on the market in the entire United States. In his testimony, Mr. Lohr said, *"Again, I don't believe that I was changing a critical component. I was replacing a burned out motor with what I had available."*⁹⁵

⁸⁸ CEMEX_NEIC_00009541-9650: 1999 RATA.

⁸⁹ CMXLYO00628865, Zurn Industries, Inc, Air Systems Div., Fan Specifications.

⁹⁰ Deposition of John Lohr on October 11, 2010, pages 60-80.

⁹¹ Deposition of Steve Goodrich on December 9, 2010, Page No. 80.

⁹² Deposition of Steve Goodrich on December 9, 2010, Page No. 79.

⁹³ Expert Report of Gerald Young, May 10, 2011.

⁹⁴ Deposition of Steve Goodrich on December 9, 2010, Page No. 80.

⁹⁵ Deposition of John Lohr on October 11, 2010, pages 60-80.

As mentioned earlier, as a part of this project neither the fan nor the fan curve was changed.⁹⁶ Also, it is important to note that the fan historically had vibration issues. The vibration occurs when over the time fan blades cake with dust, which will make fan to vibrate. However, when the fan blades are dirty, it is reasonable to assume that any increased RPMs will not proportionally relate to fan flow increase due to vibration issues.⁹⁷ It is possible that CEMEX may have to cut back to the old RMP level, to prevent vibration problems.⁹⁸

Therefore, due to other system resistance in this system, it is reasonable to assume that the minimal extra available RPMs might not have proportional 3.9% increase on the kiln flow/clinker production. Per Lohr, *“it is not proportional, because fan curves have a curve to them, so it is not liner.”*⁹⁹

Per Mr. Young’s expert report, *“when ambient air leakage and the additional steam from the water sprays used to cool the gases from 1550°F to 500°F are considered, the net available increase in the air flow rate that could support additional production is most likely in the range of 1.5 – 2.5% rather than the 3.88% of gases passing through the ID fan.”*¹⁰⁰

In a document, which was compiled by Mr. Lohr in 2000 (many years after this project), an approximate increase of 3% was assigned to this project. However, that document was based on very limited data and would not have reflected long-term variability or potential clinker production increases of this project.¹⁰¹ Review of two internal presentations indicates that this project might have a potential to increase clinker production by 28 tons per day or 9,408 tons per year.^{102,103} This presentation appear to have been compiled sometime in 2001 or 2002.

5.2.1 PSD APPLICABILITY FOR ID FAN MOTOR EMERGENCY REPLACEMENT

Project Aggregation:

In my opinion, the 1997 ID fan motor replacement should not be aggregated with the Lyons Expansion Project because it was driven by an emergency situation. Using, U.S. EPA’s justification criteria for project aggregation, there is no evidence that CEMEX included this replacement in its planning process prior to the event and capital spending decisions made subsequent to the event do not generally support reliance on the ID fan motor with respect to planned production increases in the pyroprocessing system.¹⁰⁴

⁹⁶ Deposition of John Lohr on October 11, 2010, pages 60-80.

⁹⁷ Deposition of John Lohr on October 11, 2010, pages 60-80 and Deposition of Randy Wiley on November 10, 2010.

⁹⁸ Deposition of John Lohr on October 11, 2010, pages 60-80 and Deposition of Randy Wiley on November 10, 2010.

⁹⁹ Deposition of John Lohr on October 11, 2010, pages 60-80.

¹⁰⁰ Expert Report of Gerald Young, May 10, 2011.

¹⁰¹ Deposition of John Lohr on October 11, 2010, pages 60-80.

¹⁰² CMXLYO00150942: CEMEX, Inc. Lyons Cement Plant: slide no.6. 28 tons per day equates to 9,408 tons per year clinker production increase considering 92% kiln uptime limit (8,064/8,760 hours) in the Permit No.12BO444-2 dated December 17, 1993.

¹⁰³ CEMEX_NEIC_00000203: Kiln Permit No. 12BO444-2 dated December 17, 1993.

¹⁰⁴ CEMEX_NEIC_00003286: Lyons Plant Expansion – Phase I, Project Scope, Revised December 12, 1997.

PSD Applicability for the Replacement:

CDPHE embodied the PSD regulations into Regulation 3 (5 Colorado Code of Regulations Section 1001-5). The applicable regulations under Regulation 3 were approved by the U.S. EPA. These regulations also provide an exemption for RMRR.

The U.S. EPA provides limited formal guidance as to what constitutes RMRR. From 1990 to early 1999, the U.S. EPA's RMRR guidance was largely framed by the position it took and the arguments it offered in the 1990 case involving the WEPCO.¹⁰⁵ The WEPCO case introduced a five-factor test for determining whether a routine activity could be excluded from PSD permitting.

As such, the U.S. EPA made the case that applicability of the RMRR exclusion was to be evaluated by the source on a case-by-case basis by weighing the nature, extent, purpose, frequency, and cost of the work, as well as other relevant factors to arrive at a common sense finding. U.S. EPA did not expand upon this vague definition until January 1999 as discussed above.

Depositions of CEMEX employees produced in this case point to the fact that it was a like-kind replacement. Mr. Lohr clearly mentions these facts¹⁰⁶: *"it did not occur to me that it was a notification or a permissible event. Again, I do not believe that I was changing a critical component. I was replacing a burned out motor [with] what I had available."*

Considering the like-kind replacement and uncertainty associated with PSD regulations at the time (lack of guidance on RMRR, etc.), it would have been possible to assume that this project would have fit under RMRR exclusion at the time of emergency replacement on August 8, 1997.

PSD Applicability at the Time for the Increased RPMs:

Even if this project would not have qualified as RMRR, it still would not have triggered PSD since any resulting emissions increase would have been less than significant. As discussed earlier, there might have been a minimal percent clinker production increase associated with this project. Using potential clinker production increase of 9,408 tons and average August 1995 to July 1997 baseline years' NO_x emission factor of 7.83 lb/ton (and assuming a linear relationship between production and emissions applied here), this project would have resulted in an emission increase of at most 36 tons, which is below the PSD significance threshold for NO_x emissions.¹⁰⁷ Therefore, no PSD permitting or additional analyses would have been required at that time.

Actual NO_x Emissions Before and After the Replacement:

The table below shows the actual emissions 24 month before and after this replacement. As shown, this project did not result into any NO_x increases. Please note that NO_x reductions could have been

¹⁰⁵ *Wisconsin Electric Power Co. v. Reilly*, 893 F.2d 901(7 Cir. 1990)

¹⁰⁶ Deposition of John Lohr on October 11, 2010, pages 60-80

¹⁰⁷ 9,408 tons of clinker*7.83 lb NO_x/ton of clinker (Average of 24 monthly NO_x lb/ton factors for the two baseline years prior to the project)/2000 lbs/ton= 36 tons of NO_x

possibly due to the calciner chamber and other improvement projects, which occurred prior to this project.

TABLE 5-2. ACTUAL EMISSIONS BEFORE AND AFTER THE ID FAN MOTOR REPLACEMENT

Year		Total NO_x Emissions¹ (tons)	Clinker Production² (tons)
08/1995 - 12/1995	Before the Project	697.5	184,927
01/1996 - 12/1996		1,872.5	430,272
01/1997 - 07/1997		908.0	251,995
Annual Average³		1,739.0	433,597
09/1997 - 12/1997	After the Project	642.2	165,215
01/1998 - 12/1998		1,507.3	501,008
01/1999 - 08/1999		1,297.8	320,113
Annual Average³		1,723.7	493,168
Project Increase⁴		-15.3	59,571

¹ CEMEX Lyons Facility NO_x Emissions - Kiln - Historical Summary, CMXLYO00628845.

² CEMEX Lyons Facility Estimated Clinker Production - Historical Summary, CMXLYO00628849.

³ The Annual Average is the sum of 24 month time period divided by 2.

⁴ The Project Increase is the Pre-Project Annual Average subtracted from the Post-Project Annual Average.

5.3 LYONS EXPANSION PROJECT - OXYGEN ENRICHMENT AND NO_x IMPROVEMENTS

The Lyons plant in Colorado began injecting oxygen in the kiln system experimentally in 1997 to determine the feasibility of oxygen enrichment (OE). In 1998, CEMEX management was convinced the investment in an on-site oxygen plant was desirable given the cost of purchased oxygen and the desirable clinker yields achieved during the OE experimental period.¹⁰⁸ The optimum location for the oxygen injection was determined to be near the swirl chamber located near the gas phase entrance to the pre-calciner.¹⁰⁹

CEMEX management appropriately considered whether the OE would have any effect on the pollutant emission rates from the kiln system and any associated equipment. The pollutant of concern was NO_x since OE will allow more fuel to be combusted and thus potentially generate more NO_x.

¹⁰⁸ CEMEC_NEIC_00003336: Authorization for Expenditure, February 20, 1998.

¹⁰⁹ Deposition of Steve Goodrich on December 9, 2010, page 128.

As discussed above, the formation of NO_x in a kiln system is complex. The only way CEMEX management could have determined the impact of OE on NO_x overall kiln system was to assess the actual NO_x emission rate while oxygen was injected and when it was not. Given the variable feed mix and the variety of cement products produced at the Lyons plant, the only way to effectively evaluate the impact of OE on NO_x would be to evaluate, in a statistical fashion, any small changes in NO_x emissions over a long period of time or to run a controlled short-term experiment to determine the impact of OE on NO_x.

5.3.1 1997 OE ASSESSMENT

In early 1997, CEMEX plant and corporate management planned an oxygen enrichment test to be conducted on the Lyons Plant pyroprocessing system. At the time, it was known that oxygen injected into kiln could augment fuel combustion and potentially increase production. However, there was significant concern about the aging burning zone brick and potential refractory damage associated with oxygen injection. It was decided to limit the oxygen to a maximum of 2% augmentation to avoid excessive flame temperature and refractory damage.¹¹⁰ In addition, CEMEX decided to test injection of oxygen into the calciner alone during part of the test. This was thought to be a novel approach for the industry and CEMEX was prepared to patent this element of the OE technology.¹¹¹

In February 1997, CEMEX began testing OE at the Lyons Plant. The three part test began on February 26, 1997.

1. Part 1 was designed to show the baseline operation of the kiln system. No oxygen was injected during Part 1.
2. Part 2 occurred over a fifteen day period beginning on March 3rd. Oxygen was injected into the tertiary air duct feeding the calciner.
3. Part 3 occurred over a three day period beginning on March 17th. Oxygen was injected into both the tertiary air duct and directly into the kiln with an oxygen lance.

Oxygen was injected in two separate locations – the tertiary air duct that fed directly into the combustion chamber of the calciner (Part 2 and Part 3) and directly in the kiln itself using an oxygen lance (Part 3 only). The amount of oxygen injected varied from 35,000 to 85,000 standard cubic feet per hour. The Part 3 oxygen injection rate was held constant at a rate of 70,000 standard cubic feet per hour.

Since CEMEX ultimately decided to augment combustion in the calciner only, this analysis addresses the test data collected during Part 1 and Part 2 of the 1997 OE test.¹¹²¹¹³ As shown in Figure 5-1, the

¹¹⁰ Deposition of John Lohr on October 11, 2010, Exhibit 208, CMXLYO00612385

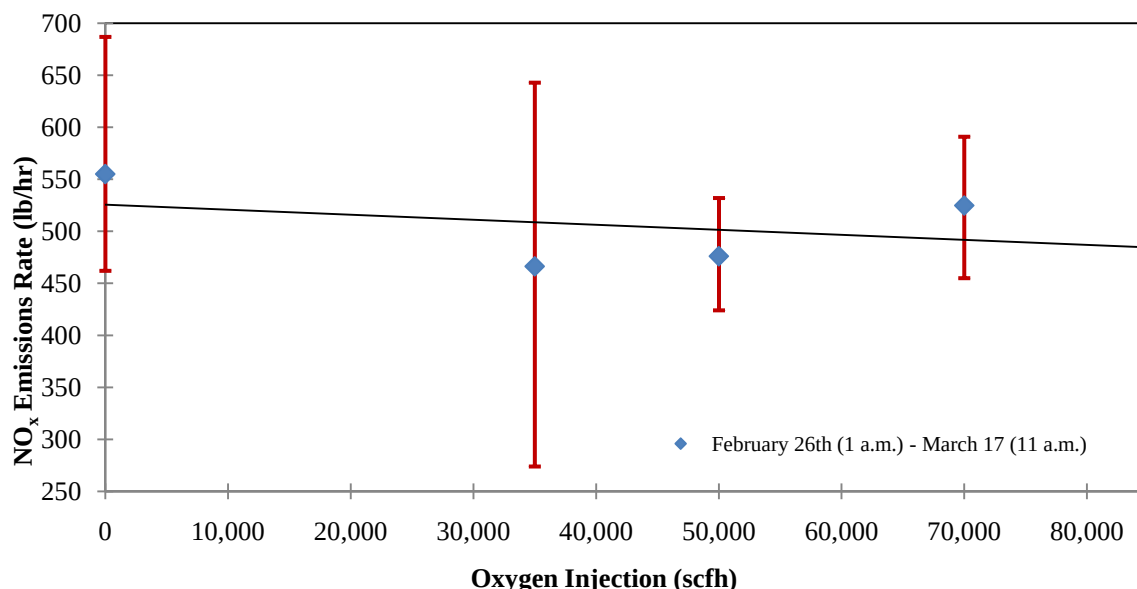
¹¹¹ Ibid

¹¹² Although not addressed in this report, it is worth noting that the analysis of the test data completed by CEMEX in 1997 indicated a general decrease in NO_x emissions during Part 3 of the test as compared to Part 1 (no OE).

¹¹³ The analysis of the 1997 OE test data contained in CMXLYO00612454 makes the assumption that the NO_x data reported under the column heading “CEM-NO Stack NO_x” are in units of pounds per hour as indicated in the deposition of Kevin Kelley on December 16, 2010 (Pages 148-149). In this dataset, I identified certain hours that appeared to be not representative of normal operations and/or resulted in “outliers” when plotted with all of the data analyzed. The hours

NO_x emission rate (pounds per hour) appears to be negatively correlated with oxygen injected into the system.¹¹⁴

FIGURE 5-1. 1997 OE TEST –PARTS 1 AND 2



As expected, the kiln feed rate and calciner coal feed rate *increased* with OE to keep the system thermally balanced and to achieve the desired production increase associated with OE. The relationships between the kiln feed rate and oxygen injection and the calciner coal feed rate and oxygen injection are shown in Figure 5-2 and Figure 5-3, respectively.¹¹⁵

excluded were: February 26 (6 pm and 8 pm), February 28 (2 pm) to March 1 (1 am) inclusive, and March 3 (12 pm and 1 pm).

¹¹⁴ Figure is prepared using data contained in CMXLYO00612454.xlsx. In this dataset, I identified certain hours that appeared to be not representative of normal operations and/or resulted in “outliers” when plotted with all of the data analyzed. The hours excluded were: February 26 (6 pm and 8 pm), February 28 (2 pm) to March 1 (1 am) inclusive, and March 3 (12 pm and 1 pm).

¹¹⁵ Ibid.

FIGURE 5-2. 1997 OE TEST –PARTS 1 AND 2 [KILN FEED VS. OE RATES]

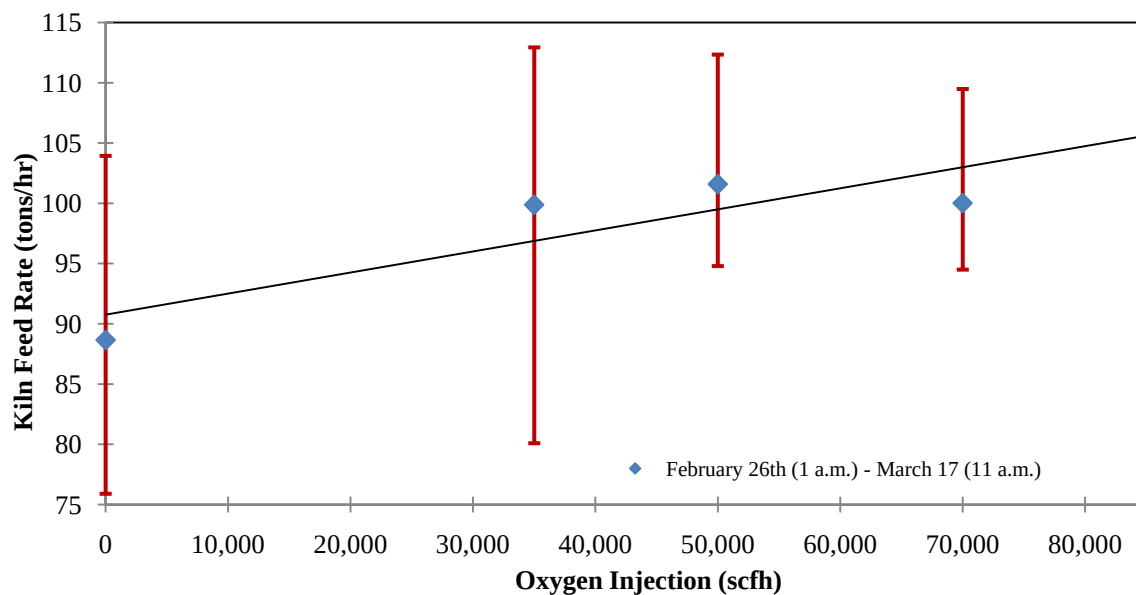
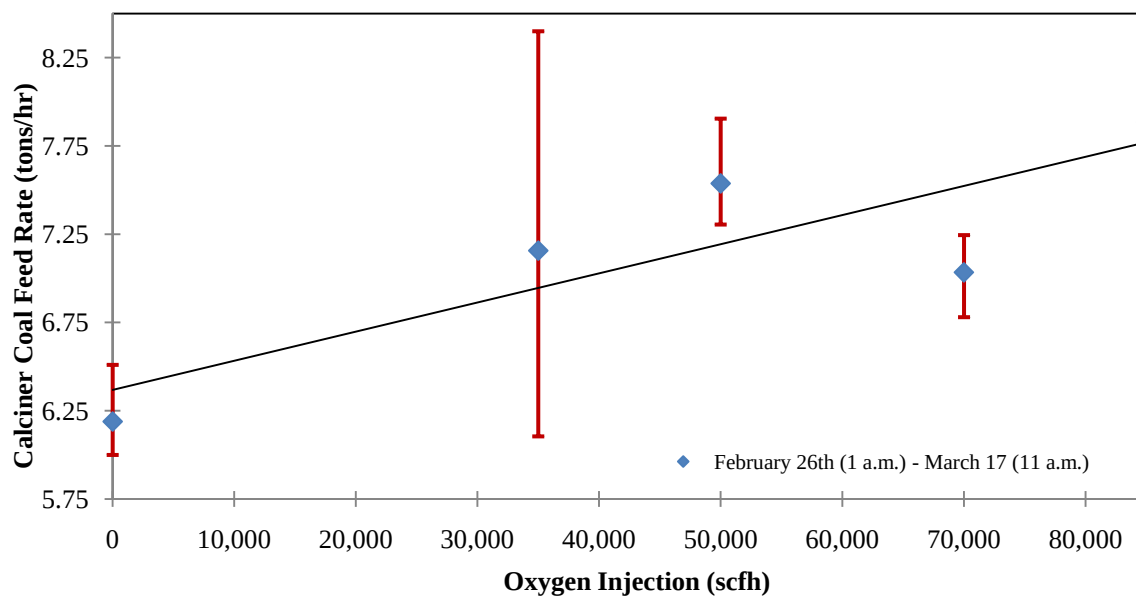


FIGURE 5-3. 1997 OE TEST –PARTS 1 AND 2 [CALCINER COAL FEED VS. OE RATES]



The data points presented in figures 5-1, 5-2, and 5-3 are average values collected at the various oxygen injection rates. The plotted bars represent the complete range of the data for each oxygen injection level.

Figure 5-4 exhibits the relationship between the calciner (pre-heater) coal feed rate during Part 1 of the 1997 OE test (no oxygen injection). As expected the NO_x emission rate increases with increased fuel consumption.¹¹⁶

FIGURE 5-4. 1997 OE TEST –PART 1 [NO_x EMISSIONS VS. CALCINER COAL FEED]

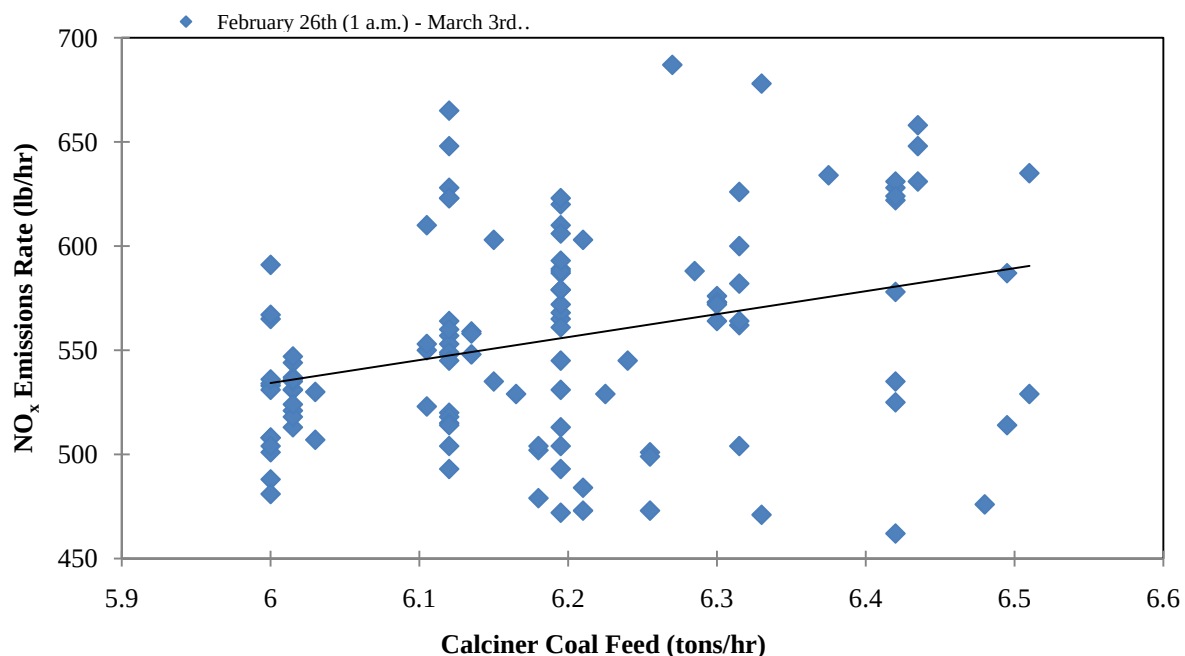
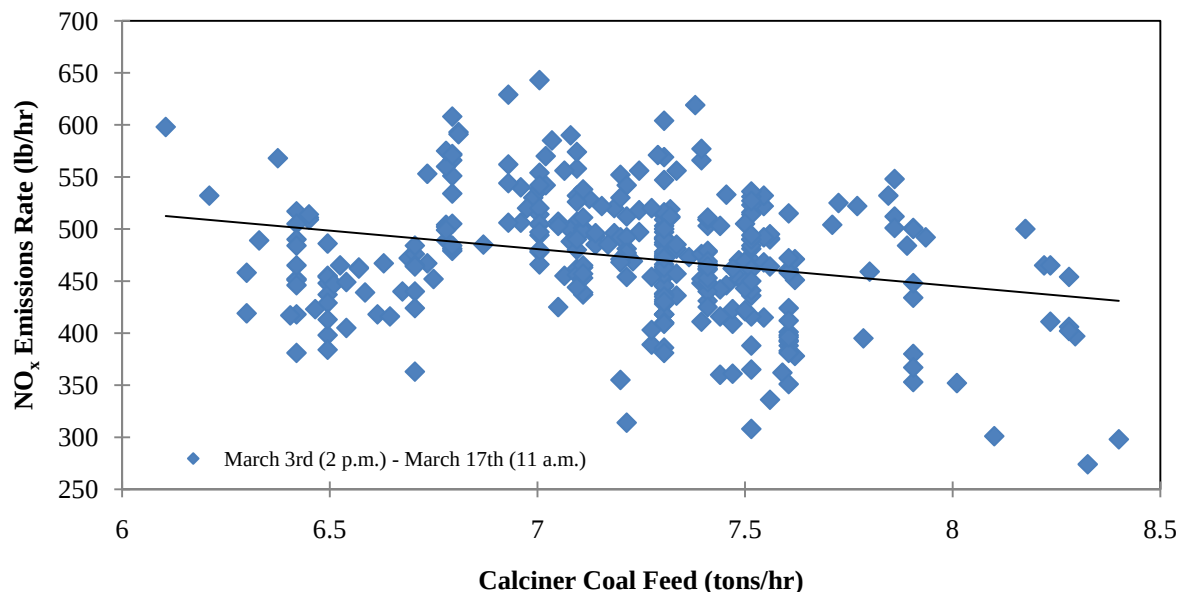


Figure 5-5 exhibits the relationship between the calciner (pre-heater) coal feed rate during Part 2 of the 1997 OE test (with oxygen at various injection rates). Note the NO_x emission rate trend is reversed as compared to Figure 5-4. With oxygen injection into the calciner, the NO_x emission rate decreases with increased fuel consumption on a mass per unit time basis.¹¹⁷

¹¹⁶ Figure is prepared using data contained in CMXLYO00612454.xlsx. In this dataset, I identified certain hours that appeared to be not representative of normal operations and/or resulted in "outliers" when plotted with all of the data analyzed. The hours excluded were: February 26 (6 pm and 8 pm), February 28 (2 pm) to March 1 (1 am) inclusive, and March 3 (12 pm and 1 pm).

¹¹⁷ Figure is prepared using data contained in CMXLYO00612454.xlsx. In this dataset, I identified certain hours that appeared to be not representative of normal operations and/or resulted in "outliers" when plotted with all of the data analyzed. The hours excluded were: February 26 (6 pm and 8 pm), February 28 (2 pm) to March 1 (1 am) inclusive, and March 3 (12 pm and 1 pm).

FIGURE 5-5. 1997 OE TEST –PART 2 [NO_x EMISSIONS VS. CALCINER COAL FEED]

The Student's t-test for Unpaired Independent Samples (t-test) is a very common mathematical method used by scientists and engineers to determine whether two populations of data (datasets) are distinct. A statistical analysis (a Student's t-test for Unpaired Independent Samples) of the 1997 OE test data was conducted to determine whether there is a statistically significant *difference* in the populations of NO_x emissions data while oxygen injection is "off" (Part 1 - OE off) and when oxygen injection is "on" (Part 2 - OE on) at any injection rate. If the two data populations are shown to be distinct, then we can be reasonably certain (95% confident) that any difference in the data means (average NO_x emissions when OE is off versus OE on) is an indication that OE influences the resulting NO_x emissions directly or indirectly.

There are two questions needed to be answered with the t-test analysis – 1) does OE decrease NO_x emissions on a pound per hour basis, and 2) does OE decrease NO_x emissions on a pound per ton of feed basis? All of the tested data sets are normally distributed. The results of the Student's t-test indicate the following:

1. Using the 95% confidence indicator, the two sets of NO_x emissions data (on a pound per hour basis) with OE on and with OE off are statistically *distinct* populations. The mean of the data with OE on is 472.9 pounds per hour. The mean of the data with OE off is 555.1 pounds per hour. OE appears to increase NO_x emissions on a pound per hour basis.
2. Using a 95% confidence indicator, the two sets of NO_x emissions data (on a pound per ton of feed basis) with OE on and with OE off are statistically *distinct* populations. The mean of the data with OE on is 4.7 pounds per ton of feed. The mean of the data with OE off is 6.3 pounds per ton of feed. OE appears to decrease NO_x emissions on a pound per ton of feed basis.

A separate statistical test was employed to assess the possible dependence of NO_x emissions on other operating parameters when oxygen is and is not injected into the tertiary air duct. The analysis of variance (ANOVA) statistical test identifies the confidence for which we can state a dataset, such as NO_x emissions (pounds per hour), is dependent on a variable, such as the coal feed rate (tons per hour) to the calciner. An ANOVA statistical test was completed on the following variables for which NO_x emissions may be dependent: a) Kiln O₂ concentration (%); b) coal feed to the calciner (tons per hour); and c) feed rate (tons per hour). The test evaluated which (if any) of these variables are likely to be indicators of NO_x emissions (pounds per hour) using the three different datasets: a) observations without OE only (Part 1 dataset), b) observations with OE only (Part 2 dataset), and c) all observations (Part 1 and Part 2 datasets combined).

The results of the ANOVA analysis suggest that the calciner coal feed rate is always a statistically significant indicator of NO_x emissions. When OE is off (Part 1 dataset), the relationship is direct (NO_x emission rate trends up with coal feed rate). When OE is on (Part 2 dataset) and when all of the data is considered (Part 1 and Part 2 datasets), the relationship is inverse (NO_x emission rate trends down when the coal feed rate increases).

The results of the ANOVA analysis suggest that the feed rate is always a statistically significant indicator of NO_x emissions with an inverse relationship in all three datasets evaluated.

The results of the ANOVA analysis suggest that the Kiln O₂ concentration is always a statistically significant indicator of NO_x emissions with a direct relationship in all three datasets evaluated.

Based on these assessments, it appears likely OE has at least some beneficial effect in reducing the amount of NO_x generated in the pyroprocessing system. In order to keep the calciner thermally balanced and to achieve the production increase associated with OE, both the kiln feed rate and the calciner coal feed rate *increase* with OE. **Despite these production related increases, the average rate of NO_x emissions on a pound per hour and on a pound per ton of feed basis were lower with OE.**

The injection of nearly pure oxygen lowers the concentration of nitrogen in the calciner and the feed and fuel based nitrogen compounds are present in slightly *greater* quantities (since OE augments the feed/fuel addition rates). We are not aware of any documented explanation of precisely how oxygen injected into the calciner reduces the overall NO_x generated in the process. A number of possibilities exist including, but certainly not limited to, the following:

- 1) A reduction in NO_x generated in the rotary kiln due to an upstream transfer of raw material calcination from the kiln to the calciner. Reducing the “load” on the rotary kiln could reduce NO_x throughout the system. As discussed above, the thermal NO_x formation process is not considered a contributor to the overall NO_x generation in the calciner. The injection of oxygen is not likely to change the combustion chamber temperature enough to change this basic assumption.
- 2) A change in the fuel rich/fuel lean zones in the calciner which will impact the combustion chemistry and potentially the net amount of NO_x generated (or reduced) in the calciner. Per Mr. Greer, “*the use of riser duct firing at Lyons has the potential to lower the NOx emission factor*

because the kiln flue gas passes through an oxygen-deficient, fuel rich zone in the riser duct, where free radicals react with and eliminate a portion of the NO_x generated in the rotary kiln.”¹¹⁸

- 3) The NO_x generated by reaction between nitrogen gas and hydrocarbon radicals in the calciner (Prompt NO_x) may be reduced because of the reduced nitrogen concentration associated with the oxygen injection. The hydrocarbon radicals are more likely to react with oxygen when oxygen is injected.

The expert report provided by Dr. Kevin A. Davis contains a detailed assessment of the possible chemical mechanisms for which NO_x could be impacted by injection of oxygen into the calciner.¹¹⁹ In addition, Dr. Davis presents NO_x emissions versus oxygen injection rate plot using some of the summarized data contained in CMXLYO00612399. The same conclusions related to the impact of oxygen injection on NO_x emissions can be drawn from these independent analyses of the 1997 OE test.¹²⁰

It is my opinion that the 1997 test provided CEMEX with a reasonable basis to conclude that OE would not increase NO_x and thus that PSD would not have been triggered.

5.3.2 1998 OE ASSESSMENT – BOARDMAN TEST

In early 1998 CEMEX engaged a statistician, Dr. Thomas Boardman, to help devise a short-term experiment as part of an ongoing effort to evaluate whether OE influenced the NO_x emission rate from the Lyons Plant pyroprocessing system. The CEMEX technical team was keenly aware of the fact that the NO_x formation chemistry in the kiln itself (primarily thermal NO_x) would not be significantly impacted by oxygen injected into the calciner. However, even slight changes in the kiln operations, unrelated to the oxygen injected during the experiment, could impact and potentially overwhelm any NO_x perturbations associated with the OE. Therefore, the kiln operations were carefully controlled for the duration of the 1998 test.

The NO_x emissions were monitored during several two hour periods when the OE was “on” and when the OE was “off” across a range of kiln excess oxygen conditions ranging from 1% to 3%. The amount of excess oxygen was controlled by the fuel firing rate. When the OE was “on”, oxygen was injected at a constant rate at the gas entrance to the calciner.

The conclusions of the Boardman Test were summarized in a letter from Mr. John Lohr (CEMEX) to Mr. Ram Seetharam (CDPHE) written on April 29, 1998.

1. NO_x emissions are not significantly influenced by the presence of OE.
2. The other criteria pollutants, SO₂, THC, & CO, show that there is no increase when OE is used.
3. There is no significant change in criteria pollutants attributable to the use of OE.

¹¹⁸ Expert Report of Walter Greer, March 1, 2011, page 23.

¹¹⁹ Expert report of Dr. Kevin Davis, May 10, 2011.

¹²⁰ Expert report of Dr. Kevin Davis, May 10, 2011.

The results of the Boardman Test are not surprising given the information obtained from the 1997 OE test and amount of NO_x emissions data collected and analyzed by CEMEX management several months leading up to the 1998 assessment. The Boardman test included very limited data and no statistical correlation was found between the formation of NO_x and the presence of injected oxygen.

Mr. Walter Greer provided an assessment of the 1998 Boardman test in his report on behalf of the United States of America.¹²¹ We find Mr. Greer's comments and overall assessment largely unfounded. The following responds to Mr. Greer's report concerning the 1998 test:¹²²

1. Greer's Comment 1: *"The fact that the test lasted for less than a day and contained ten test conditions in this short period suggests the Plant staff may have known little about the proper collection of statistical or emissions data."*

Response to Comment 1: CEMEX completed a much longer OE test in 1997 (as discussed above) and had observed NO_x emissions over the experimental period spanning several months. These data were also available to the CDPHE. The 1998 test was done primarily to address any remaining questions that may arise about the impact of Kiln O₂ (%) on NO_x emissions with and without oxygen injection.

2. Greer's Comment 2: *"The individual test conditions that lasted only two hours each would have been unlikely to produce dependable or reproducible results."*

Response to Comment 2: The air flow through the kiln system is rapid. The residence time for the gas phase NO_x reactions in the kiln system is measured in seconds. The rapid time dependent variability of the resulting short-term NO_x emissions – as pointed out by Greer - is evidence that the gas phase chemistry in the kiln system responds rapidly to changes in the key variables (oxygen concentration, flame temperature, fuel feed rate, etc.).¹²³

3. Greer's Comment 3: *"the NO_x emissions appear to increase in three out of the five conditions tested (i.e. 1.0%, 1.5%, and 2.0%). Typically, CEMEX attempted to operate its kiln so that the oxygen concentration in the flue gas at the feed end of the kiln was between 1.5% and 2.5%."*

Response to Comment 3: shows monthly average Kiln O₂ concentration as compiled from the daily operating reports produced in this case.¹²⁴ In all but five of the available data from 25 different available months evaluated, the Kiln O₂ concentration is greater than 2.5%.

¹²¹ Expert Report by Walter Greer on March 1, 2011.

¹²² Expert Report by Walter Greer on March 1, 2011.

¹²³ Deposition of Randal Wiley on November 10, 2010, Pages 73,74.

¹²⁴ References for the average monthly kiln O₂ levels are listed in the Appendix B.

TABLE 5-3. KILN O₂ MONTHLY AVERAGES

Year	Month	Kiln O₂ Average^{1,2} (%)
1997	January	2.54
	February	2.43
	March	2.23
	April	-
	May	3.19
	June	3.85
	July	3.75
	August	4.46
	September	2.88
	October	2.71
	November	3.83
	December	2.68
1998	January	4.70
	February	3.36
	March	2.25
	April	6.89
	May	4.30
	June	2.77
	July	2.35
	August	3.02
	September	2.92
	October	2.27
	November	3.20
	December	4.39
2000	January	4.86
	February	7.23

¹ The monthly average of Kiln O₂ was calculated by taking the average of the daily hourly average of the data available for the month.

² Kiln O₂ Daily Hourly Average data for 1997 - 2000 is from multiple documents summarized in the Appendix B.

4. Greer's Comment 4: *"in normal circumstances, an increase in oxygen concentration at the feed end of the rotary kiln from 1% to 3% should tend to cool the flame and concurrently reduce the NO_x generation."*

Response to Comment 4: The operational experience at the Lyons plant simply does not support Greer's statement.¹²⁵¹²⁶ The 1997 and 1998 OE test data does not support Greer's statement. Finally, the U.S. EPA's NO_x Control Document does not support Greer's statement.¹²⁷ In fact, the U.S. EPA document describes the situation in exactly the opposite fashion:¹²⁸

"The excess air used during fuel combustion can substantially affect NO formation by determining the amount of oxygen available for NO reaction. The cement kiln combustion zones usually have about 5–10% excess air, although higher excess air levels are not uncommon. Higher excess air levels result in higher NO_x."

The higher the kiln oxygen concentration, generally more NO_x will be generated within the envelope of normal operating conditions.

It is my opinion that the 1998 test along with additional supporting information available at the time provided CEMEX with support that oxygen enrichment would not result in a significant emissions increase. It was also reasonable for CEMEX to have shared its 1998 statistical analysis with CDPHE and have relied on their concurrence that permitting was not required and that this project could be permitted separately from the finish mill project.

5.4 THE LYONS EXPANSION PROJECT

Prior to this project's planning in November 1997, CEMEX experimented with the purchased liquid oxygen usage to determine exact location of oxygen injection location, injection rates, and other key variables. Through the tests conducted in 1997 and 1998, CEMEX learned that the optimum way of increasing clinker production was to inject oxygen into the swirl chamber of calciner. CEMEX started using an on-site liquid oxygen plant of 50 tons per day capacity in September 1999. Considering the timing and nature of these two projects (experimental period and on-site plant), in my analysis, we refer to these two projects as a single project – "Oxygen Enrichment (OE) project."

Documents indicate that every ton of oxygen injected resulted into an approximately 3 tons of clinker production increase.¹²⁹ The on-site oxygen plant's 50 tons capacity would have had the potential to increase approximately 150 tons of clinker production per day or approximately 50,370 tons per year.¹³⁰ Using historical yield factor of 0.59, approximately 80,000 tons of additional upstream (raw mill system) increases would have been realized. Additionally, using 5% gypsum addition in finish

¹²⁵ Deposition of Randal Wiley on November 10, 2010, Page 49.

¹²⁶ Deposition of Kevin Kelley on December 16, 2010, Page 45.

¹²⁷ David Sanders, EPA, NO_x Control Technologies for the Cement Industry (Final Report), September 19, 2000.

¹²⁸ Bill Neuffer and Mike Laney, EPA, Alternative Control Techniques Document Update – NO_x Emissions from New Cement Kilns, November 2007, Page 25.

¹²⁹ CEMEX_NEIC_00003243-3246: Lyons Kiln Production and Oxygen Use, January 4, 1998.

¹³⁰ 150 tons per day*365 days/year*92% uptime = 50,370 tons of clinker per year.

mill, 50,370 clinker increase would have been equal to an increase 52,889 tons of additional cement increases for the downstream finish mill system.

The kiln was the bottleneck throughout the history of this plant.¹³¹ In my experience with the cement plants, typically the kiln system is always the bottleneck for clinker production. In general, cement plant operator always has excess capacity in the upstream and downstream systems. Therefore, it is reasonable to believe that at the Lyons plant, the kiln is and was the bottleneck.

OE tests indicated that there were no NO_x increases from the increased production. Based on the NO_x reductions even with the increased production due to use of OE, CEMEX started planning the Lyons expansion project. Prior to the start of OE project, CEMEX conducted extensive evaluations for determining OE's impact on NO_x emissions (1997 test and 1998 Boardman test). As discussed above, these tests demonstrated that NO_x emissions most likely decreased due to the OE implementation project even with an associated production increase.

In late 1997, the Lyons plant planned a project to increase the clinker production capacity of this plant.¹³²¹³³ In addition to the oxygen enrichment in the kiln system, other purposes of this project were to modify or replace several upstream system components to achieve planned cement increase production from 525,000 tons to 600,000 tons. In addition to the oxygen enrichment project, CEMEX planned the following upstream and downstream changes¹³⁴¹³⁵:

1. Raw mill system: add a new compressor, tee damper, and replace raw mill transfer line.
2. Finish mill system: install high efficiency separator.
3. Quarry project: increase the raw material production rates.
4. Kiln system: OE and clinker cooler grates.

Consistent with U.S. EPA's criteria for the project aggregation, it is reasonable to aggregate above listed upstream and downstream projects as the Lyons Expansion Project (aggregated together as a single project for PSD applicability purposes considering U.S. EPA's aggregation guidance discussed earlier in Sections 4.1.5 and 4.1.6).

Not all of the projects mentioned in the project scoping documents, however, were implemented.¹³⁶

5.4.1 PROJECT AGGREGATION DISCUSSIONS

Based on review of the projects contemplated, initiated, and completed by CEMEX between 1997 and 2000, we have concluded that some, but not all, of these projects satisfy the aggregation criteria set forth by the U.S. EPA.

¹³¹ CEMEX_NEIC_00003243-3246: Lyons Kiln Production and Oxygen Use, January 4, 1998.

¹³² CEMEX_NEIC_00003291: Lyons Plant Expansion – Phase I, November 19, 1997.

¹³³ CEMEX_NEIC_00003300: Lyons Plant Expansion – Phase I, December 15, 1997.

¹³⁴ CEMEX_NEIC_00003291: Lyons Plant Expansion – Phase I, November 19, 1997.

¹³⁵ CEMEX_NEIC_00003300: Lyons Plant Expansion – Phase I, December 15, 1997.

¹³⁶ Deposition of John Lohr on October 11, 2010.

Table 5-4 lists the projects not associated with an overall plan to increase cement production. The table indicates the primary economic or technical driver associated with each.

TABLE 5-4. SUMMARY OF PROJECT DRIVERS AND PRIMARY IMPACTS AND PROJECTS NOT AGGREGATED WITH LYONS EXPANSION PROJECT

Project	Project Drivers and Primary Impacts
1997 Combustion Chamber Mod.	Eliminate “burnover”, efficiency improvement, reduce fuel use, reduce air pollutant emissions ¹³⁷
1997 ID Fan Motor Replacement	Unplanned emergency replacement of critical equipment ¹³⁸
2000 Annulus Boost Fan	Improve stability of kiln operations, improve product quality ¹³⁹

Overall, CEMEX management was driven to produce high quality cement in the most cost effective manner. The ID Fan motor was replaced because of a catastrophic failure of the unit. Clearly this was unplanned and could not have been reasonably anticipated.

Based on a detailed review of the testimony provided by Mr. John Lohr, all of these projects would have been implemented as stand-alone projects due to the independent manufacturing, quality aspects, and fuel use/cost benefits associated with each.¹⁴⁰ There is no indication that CEMEX senior management required Mr. Lohr to justify these projects as a group or as part of an overall plant expansion plan. In fact, there is ample evidence these projects were independently conceived and implemented.

The calciner combustion chamber was modified in 1997 to rectify significant problems with the unit design. Due to insufficient fuel and air mixing characteristics in the combustion chamber, the calciner was extremely inefficient resulting in the following:

1. poor fuel utilization,
2. “burnover” outside the combustion chamber,
3. periodically high CO emissions,
4. poor heat transfer characteristics,
5. reduced calcination, and
6. clinker quality issues.

The Annulus Boost fan project occurred nearly three years after the projects primarily associated the Lyons Expansion Project. The new Annulus Boost fan improved the combustion characteristics in the kiln leading to product quality improvement and potentially a modest increase in clinker production. The timing and nature of the Annulus Boost fan project; however, does not satisfy the US EPA’s 1989 and 1993 aggregation policy guidance with respect to the earlier projects.

¹³⁷ Deposition of John Lohr on October 11, 2010.

¹³⁸ Deposition of John Lohr on October 11, 2010.

¹³⁹ Deposition of Steve Goodrich on December 9, 2010.

¹⁴⁰ Deposition of John Lohr on October 11, 2010.

Table 5-5 lists the components of the Lyons plant projects. These projects appear to satisfy some of the U.S. EPA's aggregation criteria considering their nature, timing, and context under which they were conceived. The internal correspondence we have reviewed appears to indicate CEMEX management considered these projects integral to the ability of the plant to increase clinker and cement production to meet market demand.

TABLE 5-5. SUMMARY OF PROJECT DRIVERS AND PRIMARY IMPACTS FOR LYONS EXPANSION PROJECTS

Project	Project Drivers and Primary Impacts
Oxygen Enrichment/VSA Plant	Increase clinker production. ¹⁴¹¹⁴²
Finish Mill - High Efficiency Separator	Replace aging equipment, increase clinker grinding capacity ¹⁴³
Raw Mill – Transfer Line	Replace exiting line, increase raw material Transfer
Quarry Project	Increased raw material
Raw Mill – Compressor	Debottleneck raw mill ¹⁴⁴
Clinker Cooler upgrades	Improve cooling efficiency ¹⁴⁵¹⁴⁶

Concerning U.S. EPA's Criteria 4 in the June 13, 1989, memorandum,¹⁴⁷ CEMEX management presented detailed project information to the CDPHE in the form of a PowerPoint presentation on March 6, 1998.¹⁴⁸ Nowhere in this presentation is there any evidence that CEMEX management made "statements ... to state or local permitting agencies about the source's plans for operation ... show[ing] intent to circumvent preconstruction review requirements." An informal memorandum composed by Mr. Dennis Myers of the CDPHE during the same time-frame makes clear the CDPHE was alert to the possibility of *potential* project aggregation depending on how CEMEX proceeded with subsequent permitting.¹⁴⁹ Therefore, it appears unlikely the CDPHE would have ignored the issue in a contemporaneous or prospective analysis of the projects presented by CEMEX.

In summary, the Lyons plant projects can be aggregated using the criteria described in the U.S. EPA's policy documents available at the time. There are only tenuous, at best, technical and economic ties between these projects and the others evaluated in this report. A more appropriate assessment of the project planning and decision making at the time clearly points to CEMEX plant management dealing with largely isolated operational challenges prior to late 1997 as they occurred and funding was available to address them. There is no evidence that CEMEX intentionally altered the timing, project descriptions, or planning process in an attempt to disaggregate and therefore avoid major NSR.

¹⁴¹ Deposition of John Lohr on October 11, 2010.

¹⁴² Deposition of Steve Goodrich on December 9, 2010, page 24.

¹⁴³ Deposition of Steve Goodrich on December 9, 2010, page 145.

¹⁴⁴ CEMEX_NEIC_00008527

¹⁴⁵ CEMEX_NEIC_00003295

¹⁴⁶ Deposition of John Lohr on October 11, 2010, pages 158-160.

¹⁴⁷ Terrell E. Hunt and John S. Seitz, U.S. EPA, Guidance on Limiting Potential to Emit in New Source

Permitting, June 13, 1989.

¹⁴⁸ US00000594 to US00000605

¹⁴⁹ US00006499

5.4.2 DEBOTTLENECK DISCUSSIONS FOR UPSTREAM RAW MILL SYSTEM

The 1993 kiln permit's NO_x enforceable limit was derived using the following basis:¹⁵⁰

- Kiln feed rate: hourly average 110 tph, hourly maximum 120 tph, and annual 887,040 tons.

Review of multiple documents indicates that indicate the raw mill capacity has always been higher than the kiln's maximum feed rate of 120 tph, indicating that the kiln was the bottleneck. As discussed earlier, due to OE, it was possible to increase the kiln feed rate/clinker production rate without any NO_x increases. This increased kiln feed rate would have required additional output from the raw mill system as a part of the Lyons Expansion Project and the available excess raw mill capacity might have been close to the increased kiln feed rate capacity.

Per Mr. Lohr, *"the improvements in kiln operation in 1997 reversed that situation. With the combustion chamber improvements and the injection of liquid oxygen, the kiln began to outrun the upstream and downstream systems."*¹⁵¹ In order to ensure that that the raw mill continue having excess production capacity (this is how typically cement plants are operated), as a part of the Lyons Expansion project, CEMEX management installed an additional compressor and replaced the transfer line. Also, per Mr. Lohr's deposition, some of these changes were necessary to fix line plugging problems.

Per Mr. Young's expert report, *"it is also noted that the raw mill system and the pyroprocessing system are not directly connected. There is a homogenizing silo and a kiln feed silo that each provides surge capacity for raw mill product storage."*¹⁵²

Therefore, these upstream raw mill changes, which were made as a part of the LEP, should not be considered as the debottleneck changes. Even after these changes, the kiln operating at 120 tph (lower capacity than 150 tph raw mill capacity) continued to be the bottleneck on overall production at the plant.

The upstream raw mill system only contributes to particulate matter emissions; it does not contribute any NO_x emissions. Since the kiln's increased feed/clinker capacity was only achievable through OE injection, which resulted in no increase in NO_x emissions, these upstream changes would not have impacted kiln NO_x emissions.

CEMEX started the necessary permitting process for with the CDPHE as early as in January 1998.¹⁵³ Since these raw mill changes would have only impacted the PM₁₀ emission levels, necessary discussions and approvals were obtained.

¹⁵⁰ CEMEX_NEIC_00000205: 1993 Kiln Permit.

¹⁵¹ CEMEX_NEIC_00003243-3246: Lyons Kiln Production and Oxygen Use, January 4, 1998.

¹⁵² Expert report of Gerald Young, May 10, 2011.

¹⁵³ US00006500: Application process history.

5.4.3 PSD APPLICABILITY AT THE TIME

Oxygen enrichment was the key to achieving planned clinker production increases at Lyons Plant. Experiments demonstrated that the OE injection did not contribute to any NO_x emission increases but had capability of achieving clinker production increases. As mentioned earlier in my report, despite the production related increases, the average rate of NO_x emissions on a pound per hour and on a pound per ton of feed basis were lower with OE.

Per Mr. Goodrich's deposition, the OE project would not have any impact on the kiln system's reliability but would have improved the system stability.¹⁵⁴ In general, reliability is a measure of the uptime and stability is a measure of efficiency. Improved stability should improve efficiency/reduce NO_x emissions.^{155,156} Review of the kiln system's uptime data indicate that kiln's uptime performance did not significantly change during the 1996-2001 period. Therefore, this project would not have resulted into any additional clinker production increases due to reliability reasons, which is consistent with Mr. Goodrich's deposition.

TABLE 5-6. SUMMARY OF KILN'S OPERATING HOURS

Year	Kiln Hours of Operation ¹ (hrs/yr)	Kiln Uptime (%)
1995	7,962	90.9
1996	7,969	91.0
1997	7,759	88.6
1998	7,974	91.0
1999	7,953	90.8
2000	8,020	91.6
2001	7,923	90.4
2002	7,526	85.9
2003	7,871	89.9
2004	7,927	90.5
2005	7,641	87.2
2006	7,921	90.4

¹ From CMXLYO00113389-409 (native).xls.

Injection of OE may be considered as a "change in method of operation." However, 1997 and 1998 test data show that the planned clinker production would not have resulted into any NO_x emission increases. Again,

¹⁵⁴ Deposition of Steve Goodrich, December 9, 2010, page 37.

¹⁵⁵ Deposition of Steve Goodrich, December 9, 2010, page 37.

¹⁵⁶ Expert Report of Gerald Young, May 10, 2011.

"The NSR regulatory provisions require that the physical or operational change "result in" an increase in actual emissions in order to consider that change to be a modification (see, e.g., 40 C.F.R. 52.21(b)(2)(i))."

In other words, as discussed earlier in Section 4.1.4, PSD does not apply unless the source finds that there is a causal link between the proposed change and any post-change increase in emissions. Therefore, it would have been reasonable to assume that no PSD permitting was needed for the Lyons Expansion Project. CEMEX confirmed this approach with CDPHE.

Since there were no NO_x increases, this project would not have met the "modification" definition under PSD regulations effective at that time. Therefore, it is reasonable to assume that no PSD would have applied back in early 1998 when this project was planned.

CEMEX discussed the Lyons Expansion Project with CDPHE, including on March 6, 1998, and explained that the overall NO_x emissions will either remain unchanged or decreased as a part of this project.¹⁵⁷ Since this project did not show any NO_x emissions increases, no permitting was done for the kiln production increase project.

5.4.4 MMA'S APPROACH ON LYONS'S EXPANSION PROJECT

In order to evaluate the impacts of these planned projects and determine PSD applicability, CEMEX hired MMA.

On March 25, 1998, Mr. Keith Baugues of MMA drafted a letter to Mr. Gary Parish of LeBoeuf, Lamb, Green & MacRae describing proposed plant and quarry projects, summarizing various permitting considerations and related matters associated with plant expansion options being contemplated by CEMEX management. This letter addressed the following changes contemplated at the time it was written:

1. The addition of a new high pressure separator
2. An increase in the output from the Dowe Flats quarry
3. An increase in the material transported from the Dowe Flats quarry to the plant
4. An increase in the annual throughput at the plant kiln specifically pertaining to oxygen enrichment (OE).

This letter was written approximately two weeks after Mr. Baugues sent a letter to Mr. John Lohr (dated March 12, 1998) addressing some of the same permitting considerations but without specifically delineating the projects noted in the March 25th letter.

In the March 25th letter, Mr. Baugues correctly illuminated a key issue with respect to PSD permitting applicability. He notes that PSD permitting is not required for the kiln system oxygen injection

¹⁵⁷ US00000594: Southdown, Inc., Lyons Plant, Meeting with CDPHE on March 6, 1998.

project if certain “requirements” are met. One of these requirements is the absence of any emissions increases associated with OE.

The March 12th and March 25th letters represent a natural progression in Mr. Baugues’ assessment of the expansion plans at the Lyons plant. An air quality consultant will often develop a rough estimation of the project emissions change and communicate these to the client. This was the focus on the March 12th letter. The March 25th letter offered a more detailed analysis of the projects, estimated emissions, and provided CEMEX management with a basis to lawfully and efficiently address the regulatory requirements and possible exemptions associated with the permitting of the proposed projects. Specifically, this letter appears to be an impetus for plant management to proceed with the 1998 Boardman OE test to document what they already knew about OE, specifically OE’s dampening effect on kiln system NO_x and other pollutant emissions.

5.4.5 ACTUAL NO_x EMISSIONS BEFORE AND AFTER THE LYONS EXPANSION PROJECT

Table 5-7 shows the actual NO_x emissions before and after the Lyons Expansion Project. As discussed above, the US EPA revised its NSR regulations in 2002 to allow facilities to utilize an “actual-to-projected-actual” methodology for determining applicability (67 FR 80,186). Colorado adopted similar revisions on April 16, 2004. Under this methodology, CEMEX could have concluded that PSD was not triggered since, for the reasons discussed elsewhere in this report, it had reasonable bases for projecting that actual emissions following and resulting from the projects would not have increased, let alone significantly.

TABLE 5-7. ACTUAL NO_x EMISSIONS BEFORE AND AFTER THE LEP

Periods	Clinker Production¹ (tons)	CKD Free Clinker Production^{2,3} (tons)	NO_x Emissions⁴ (tons)
Pre-Project (24 Months Prior to September 1997) ⁵	435,160	413,667	1,753.4
Post Project (24 Months After May 1999) ⁵	523,304	457,818	1,787.7
Project Increase⁶	88,144	44,151	34.4

¹ CEMEX Lyons Facility Estimated Clinker Production - Historical Summary, CMXLYO00628849.

² CEMEX_NEIC 8531.xls.

³ Cement Kiln Dust Return Data is not available for November 1995 and January - September of 1996.

⁴ CEMEX Lyons Facility NO_x Emissions - Kiln - Historical Summary, CMXLYO00628845.

⁵ Values shown are the sum of the 24 month time period divided by 2.

⁶ Project Increase (tons) = Post Project (tons) - Pre-Project (tons)

5.4.6 CKD RECYCLING

The Lyons plant recycles CKD. Adjusting the total clinker production numbers with CKD free clinker production, it appears that clinker production almost remained constant during the alleged Lyons Expansion Project period.

Alkali content in the portland cement is dependent on type of cement being produced. For some type of cement such as low-alkali cement, alkali content is intentionally kept very low. Alkali bypass (ABP) unit at the Lyons Plant is designed to remove a portion of kiln flue gas to maintain desirable alkali content. Removed gas in ABP are cooled and solids material collected in the ABP baghouse are considered as cement kiln dust (CKD). In addition, similar CKD losses are collected in the main kiln baghouse. CKD is a material that gone through significant calcination process in the kiln system. Therefore, it is not prudent to dispose this material. Plants try to recycle and or use CKD. The amount of CKD recycled back into the kiln system as a raw material depends on the type of cement produced and alkali content requirement.

In the Mid 1980s, significant amount of CKD was disposed and wasted due to high demand for Type II low alkali cement.¹⁵⁸ As a result, the Lyons plant wasted approximately 100 tons per day or 34,000 tons of CKD per year from ABP baghouse and 50 tons per day or 17,000 tons of CKD from main kiln baghouse resulting in approximately 51,000 tons of CKD per year.¹⁵⁹ With the introduction of Dowe Flats quarry raw material to the kiln, the plant was able to improve the raw mix and significantly reduce CKD wastage.¹⁶⁰ Over the years, the Lyons Plant has improved its performance on CKD recycling. The 1998 operating year's performance summary clearly mentions that with the implementation of the mix optimization program, CKD waste was minimized resulting in increased production.¹⁶¹

Per Mr. Greer, *"the CKD at Lyons is calcined to a significant degree, e.g., 75%. Calcination is the most energy intensive step in the pyroprocess; therefore, less fuel is required to convert CKD to clinker than is required for clinkering virgin raw mix. Also, less gas is generated from calcining and clinkering CKD, so more of the gas handling capacity of the ID fan is available for increasing clinker production."*¹⁶²

In general, production capacity of a cement kiln is determined by subtracting CKD removed from the total kiln feed or alternatively from the produced clinker. Table 5-8 below shows total clinker produced, CKD recycled back into the kiln, and CKD adjusted total clinker production for the Lyons plant.

¹⁵⁸ CMXLYO00045614: Kiln Volatile Management prepared by Royce Smith in March 20, 2000.

¹⁵⁹ CMXLYO00155868: O₂ Enrichment Test, Lyons Colorado Plant, March 1997.

¹⁶⁰ CMXLYO00045614: Kiln Volatile Management prepared by Royce Smith in March 20, 2000.

¹⁶¹ CMXLYO00045794: Lyons Plant Performance Summary – 1998 Operating Year.

¹⁶² Expert Report by Walter Greer on March 1, 2011, Page No. 17.

TABLE 5-8. TOTAL CLINKER AND CKD-FREE CLINKER PRODUCTION LEVELS

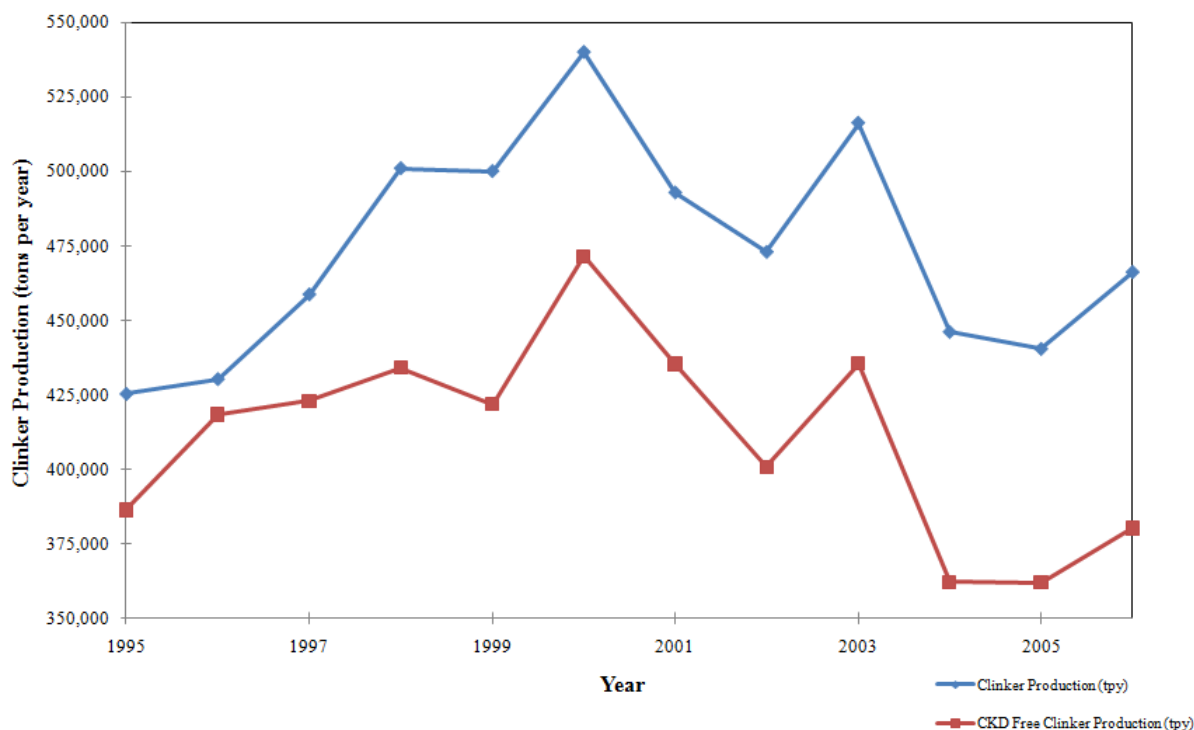
Year	Estimated Clinker Production¹ (tons)	Cement Kiln Dust Return² (tons)	CKD Free Clinker Production³ (tons)
1995	425,274	38,932	386,343
1996	430,272	11,838	418,434
1997	458,694	35,737	422,957
1998	501,008	66,901	434,107
1999	500,031	78,099	421,933
2000	539,992	68,371	471,621
2001	492,855	57,381	435,474
2002	472,945	72,041	400,904
2003	516,251	80,753	435,498
2004	446,097	83,967	362,130
2005	440,383	78,401	361,982

¹ Clinker Production for 1995-2005 is from CEMEX Lyons Facility Estimated Clinker Production - Historical Summary, CMXLYO00628849.

² Cement Kiln Dust Return is from CEMEX_NEIC 8531.xls.

³ CKD Free Clinker Production (tpy) = Estimated Clinker Production (tpy) - Cement Kiln Dust Returned (tpy)

FIGURE 5-6. CLINKER AND CKD FREE CLINKER PRODUCTION LEVELS



As shown above in the table and graph, CKD adjusted maximum clinker production of 471,621 tons was observed in 2000. For all other years, it remained below 435,498 tons, well below the historical capacity of this kiln. Increased CKD recycling would beneficially impact NO_x emissions of the system (i.e. lower NO_x emissions per ton of clinker produced).

5.5 ANNULUS BOOST FAN

In 2000, an annulus boost fan was installed.¹⁶³ Although a similar project was originally contemplated to be included within the scope of the LEP, it appears that the facility abandoned the project in 1998. It further appears that the installation of the fan in 2000 was an independent decision and distinct from the LEP. For these reasons, it is my opinion that this project need not be aggregated for PSD purposes with other projects undertaken at the facility.

This fan was installed to improve the flame characteristics in the burning zone by increasing the velocity of the coal and primary combustion air at the tip of multi-channel burner pipe. Overall, the goal was to improve the stability of the system and provide better combustion.¹⁶⁴ Mr. Gerald L. Young evaluated this project in his expert report developed for this case. In his expert opinion, the boost fan would have improved kiln operational stability, which in turn would have reduced NO_x.

Regardless of whether the project is evaluated independently or aggregated with the LEP, it would not have subjected the plant to PSD. Assuming Mr. Young's assessment to be true, the project would not have resulted in a significant net emissions increase if analyzed alone or contributed to a significant increase if aggregated with the LEP.

5.6 USE OF OTHER ALTERNATE RAW MATERIALS

Based on Mr. John Lohr's deposition exhibit 201, the following alternate raw materials related changes may have also occurred at the Lyons Plant during the 1997-2000 time frame.

- Fly ash addition
- Steel slag injection on feedshelf
- Steel slag injection at raw mill

The depositions of Mr. Lohr and Mr. Goodrich indicate that the fly ash addition was implemented in 1998 and discontinued shortly after.^{165,166} Fly ash is considered as alternate raw material. Addition of the fly ash is part of the raw material mix substitution at any cement plant. For multiple reasons, each cement plant continuously evaluates the raw material mix design.

Per Mr. Lohr, steel slag injection on the feedshelf was aborted within a year. Also, steel slag injection at the raw mill failed due to the material being too hard for the raw mill.¹⁶⁷ Mr. Goodrich

¹⁶³ CMXLYO00157315: Lyons Kiln Status, August 2000.

¹⁶⁴ Expert Report of Mr. Gerald Young, May 10, 2011.

¹⁶⁵ Deposition of Mr. John Lohr, October 11, 2010.

¹⁶⁶ Deposition of Mr. Steve Goodrich, December 9, 2010.

¹⁶⁷ Deposition of Mr. John Lohr, October 11, 2010.

also confirmed that steel slag injection only lasted for a few months in early 2000.¹⁶⁸ Similar to the fly ash, steel slag addition is also considered as a raw material mix substitution.

These raw material usages are exempt from PSD. PSD regulations provide specific exclusion for the raw material/fuel usage from the Regulation 3 definition of major modification. These are:

“Use of an alternative fuel or raw material by a stationary source which, prior to January 6, 1975 for a source in an attainment or unclassifiable area, and prior to December 21, 1976 for a source in a nonattainment area, the stationary source was capable of accommodating or the source is approved to use under any permit issued under 40 CFR 51 [Subpart I] or 52.21 or under Regulation 3.”

Therefore, the raw material related changes at Lyons Plant would be exempt from the PSD reviews. Additionally, these changes itself are not related to the LEP. In general, these changes also do not change NO_x emissions level of the kiln. Similar to CKD, Slag addition might even provide NO_x improvements in absolute terms and/or on a pound per ton of clinker basis.

¹⁶⁸ Deposition of Mr. Steve Goodrich, December 9, 2010.

6. BACT/LAER/ECONOMIC BENEFIT DISCUSSIONS

This section discusses our review and analysis of what likely would have constituted Best Available Control Technology (BACT) and Lowest Achievable Emission Rate (LAER) at the facility between 1997 and 2000. This section also critiques the opinions of Dr. Ranajit Sahu, the United States expert on this topic.

A requirement of PSD permitting is the establishment of an emission limit achievable through the use of BACT. The Clean Air Act section 169 defines BACT as:¹⁶⁹

an emissions limitation (including a visible emission standard) based on the maximum degree of reduction for each pollutant subject to regulation under Act which would be emitted from any proposed major stationary source or major modification which the Administrator, on a case-by-case basis, taking into account energy, environmental, and economic impacts and other costs, determines is achievable for such source or modification through application of production processes or available methods, systems, and techniques, including fuel cleaning or treatment or innovative fuel combustion techniques for control of such pollutant. In no event shall application of best available control technology result in emissions of any pollutant which would exceed the emissions allowed by any applicable standard under 40 CFR parts 60 and 61. If the Administrator determines that technological or economic limitations on the application of measurement methodology to a particular emissions unit would make the imposition of an emissions standard infeasible, a design, equipment, work practice, operational standard, or combination thereof, may be prescribed instead to satisfy the requirement for the application of best available control technology. Such standard shall, to the degree possible, set forth the emissions reduction achievable by implementation of such design, equipment, work practice or operation, and shall provide for compliance by means which achieve equivalent results.

Colorado's regulations define BACT in an almost identical way.

BACT is determined on a project-by-project and pollutant-by-pollutant basis using the criteria set forth in the Clean Air Act and EPA guidance. See U.S. EPA's *NSR Workshop Manual*. U.S. EPA guidance recommends that facilities/permitting authorities follow a six step process for determining BACT:

- Step 1: Identify all control technologies
- Step 2: Eliminate technically infeasible options
- Step 3: Rank remaining control technologies by control effectiveness
- Step 4: Evaluate most effective controls and document results
- Step 5: Select BACT

Among other factors, economic costs, environmental impacts, feasibility, and performance must be taken into account in determining a BACT limitation. U.S. EPA and the states maintain a

¹⁶⁹ 40 CFR 52.21(b)(12)

clearinghouse of BACT determinations in a database known as the RACT/BACT/LAER (RBLC) Database.¹⁷⁰

In his report, Dr. Sahu suggests that NO_x BACT for the Lyons plant would be Selective Catalytic Reduction (SCR), unless “a detailed cost analysis deem SCR not to be cost-effective,” in which case “BACT for this plant would be a combination of the following, if they are deemed cost effective . . . (a) indirect firing and LNB [low-NO_x burners], with a NO_x reduction efficiency of 30% (conservatively); and (b) SNCR [selective non-catalytic reduction] with a NO_x reduction of 65%.”

It is my opinion that Dr. Sahu’s conclusions are without support and that his analysis is admittedly flawed. Dr. Sahu himself acknowledges that he did “*not conduct a complete BACT analysis*,” and that he did not consider cost, a statutorily required criterion for determining BACT. Consequently, he has no justifications for his conclusion other than the fact that indirect firing, LNB, SCR, and SNCR are technically feasible today. This is not, however, sufficient for BACT purposes.

Dr. Sahu’s suggestion that these technologies and an emission level of 75.5% percent would have been deemed BACT “at all times since 1996” is inconsistent with BACT determinations made between 1997 and 2000, as inventoried on U.S. EPA’s RBLC database; BACT determinations CDPHE and EPA Region 8 made in 1999 and 2003, respectively; and U.S. EPA’s recent New Source Performance Standards for cement kilns.

According to U.S. EPA’s RBLC, between 1997 and 2000, 13 portland cement facilities underwent BACT review. See Table 6-1. Of those, *none* was required to install add-on pollution controls for NO_x. Add-on controls for NO_x (e.g., SNCR) did not start to become prevalent until late 2004. Prior to 2002, there was only one facility, Great Star Cement, that agreed to install SNCR, but that kiln was never constructed. As a result, that determination has been universally ignored or dismissed by state permitting authorities. The first time a domestic facility installed SNCR was in 2002. That facility (Continental Cement), however, had a NO_x emission limit of 8 lbs per ton of clinker, which is far higher than the limit suggested by Dr. Sahu. The first kiln actually built to receive a PSD permit with a SNCR-based NO_x limit was Lehigh Cement in December 2003. This information refutes Dr. Sahu’s report.

In 1999, CDPHE permitted a “greenfield” cement plant in Colorado and did not require SNCR or any other add-on control device as BACT. Another PSD permitting project in U.S. EPA Region 8 done later in 2003 also did not consider SNCR as a BACT. Almost without exception, NO_x BACT determination made during this period relied on “good combustion controls” or inherent kiln design.

In his report, Dr. Sahu states a NO_x BACT limit would correlate with an emission limit of 1.31 lbs/ton of clinker. Again, Dr. Sahu has no support for this assertion. Moreover, it is also inconsistent with U.S. EPA’s recently promulgated NSPS for portland cement kilns [75 Fed. Reg. 54,970 (September 9, 2010)]. That rule establishes a limit of 1.5 lbs/ton of clinker based on the assumptions that a new preheater/precalciner kiln would have a NO_x baseline of 3.0 lbs/ton of clinker and that SNCR

¹⁷⁰ See <http://cfpub.epa.gov/RBLC/index.cfm?action=Home.Home&lang=en>.

technology has an efficiency of 50 percent. U.S. EPA acknowledges, however, that this limit is not necessarily feasible for existing kilns, like Lyons, with NO_x baselines above 3.0 lbs/ton of clinker.

I also disagree with Dr. Sahu's assertion that indirect firing would have constituted BACT in 1996 and later. In my experience, there has not been a PSD project where indirect firing alone would have been considered as an add-on control system for PSD BACT. Typically, owners of new or modified kilns, as a part of the basic kiln design, consider installing the indirect firing system. The Lyons kiln already provides the inherent NO_x reduction due to pre-heater/pre-calciner kiln design.

Based on my evaluation of the BACT determinations made during the relevant period, it is my opinion that had CEMEX undergone PSD review between 1997 and 2000, it is unlikely that CDPHE would have required the plant to install add-on controls or to modify the kiln design in any significant way (e.g., indirect firing). It is likely that CDPHE would have set the NO_x BACT limit at a level consistent with "good combustion practices" at the plant.

An estimation of BACT as of 2011 would require a full top-down BACT analysis. However, the most likely BACT technology would be SNCR.

In my experience with the U.S. EPA's nonattainment NSR permitting requirements, LAER is only triggered when a project causes a nonattainment pollutant to increase significantly, here PM₁₀. Because no such increase occurred at the Lyons plant because of these projects, LAER would not be required. And even if it were, I believe that the baghouse currently employed at the Lyons plant would have met the LAER requirements applicable at that time.

Because it is my opinion that had CEMEX undergone BACT review between 1997 and 2000, no add-on controls would have been required, it is also my opinion that CEMEX did not economically benefit by avoiding PSD permitting.

Dr. Sahu's opinion that CEMEX experienced a \$44.7 million benefit is fundamentally flawed and unsupportable. First, he incorrectly assumes that by avoiding PSD permitting, CEMEX avoided a \$7 million capital cost for indirect firing and a \$600,000 cost for SNCR. As discussed above, neither of those technologies likely would have constituted BACT between 1997 and 2000. Second, Dr. Sahu incorrectly calculates noncompliance over 12 years (between 1997 and 2012). It is my experience that U.S. EPA calculates noncompliance only over the 5-year statute of limitations period. Dr. Sahu's estimates are also inconsistent, by many orders of magnitude, with U.S. EPA's penalty calculations in other PSD cases.¹⁷¹

¹⁷¹ See <http://cfpub.epa.gov/compliance/cases/>.

Table 6-1. RBLC NO_x Control and Emission Rate Determinations From 1997-2000 ¹

RBLC ID	Facility Name	Facility State	Permit Number	Application Accepted Received Date ²	Permit Issuance Date	Process Name	Control Method Description ²	NO _x Emission Limit 1 ²	NO _x Emission Limit 1 Unit	NO _x Emission Limit 2 ²	NO _x Emission Limit 2 Unit ²	NO _x lb/ton Clinker
CO-0043	RIO GRANDE PORTLAND CEMENT CORP.	CO	98PB0893	12/30/1998 ACT	09/25/2000 ACT	PREHEATER/PRECALCINER, KILN	MULTI-STAGE COMBUSTION AND RECIRCULATION. EMISSION LIMIT IN LB/T OF CLINKER. LB/T LIMIT IS 12-MO ROLLING AVG.	2.32	LB/T	1,100.00	T/YR	2.32
CO-0047	HOLNAM, FLORENCE	CO	98-FR-0895	-	07/29/1999 ACT	KILN/PREHEATER/BYPASS & CLINKER COOLER EXHAUST	LOW NOX COMBUSTION SYSTEM	2,922.71	T/YR	-	-	Cannot be determined from RBLC data.
CO-0048	HOLNAM, LAPORTE CO.	CO	11LR338-1	02/02/1998 ACT	09/22/1998 ACT	CALCINER/ KILN	SPECIAL PROCESS: DESIGN OF BURNER/KILN TO CONTROL ALKALI FROM LIMESTONE INCREASES NOX EMISSIONS.	900.00	LB/H	2,500.00	T/YR	Cannot be determined from RBLC data. ³
FL-0139	SUWANNEE AMERICAN CEMENT COMPANY, INC.	FL	1210465-001-AC	11/30/1999 ACT	06/01/2000 ACT	IN LINE KILN & RAW MILL	MULTI-STAGE COMBUSTION W/SEPARATE LINE CALCINER COMBUSTION CHAMBER.	2.90	LB/T	-	-	2.90
FL-0173	SOUTHDOWN, INC.	FL	PSD-FL-233	02/22/1996 ACT	06/27/1997 ACT	PORTLAND CEMENT, KILN 2	GOOD COMBUSTION PRACTICES AND BURNER DESIGN WITH PRIMARY COMBUSTION AIR CONTROL.	1.72	LB/T	-	-	Cannot be determined from RBLC data. ⁴
FL-0173	SOUTHDOWN, INC.	FL	PSD-FL-233	02/22/1996 ACT	06/27/1997 ACT	PORTLAND CEMENT, KILN 1	GOOD COMBUSTION PRACTICES AND BURNER DESIGN WITH PRIMARY COMBUSTION AIR CONTROL.	1.80	LB/T	-	-	Cannot be determined from RBLC data. ⁴
IL-0057	ILLINOIS CEMENT COMPANY	IL	97030016	03/03/1997 ACT	06/12/1998 ACT	KILN, CEMENT, PREHEATER-PRECALCINER	CONVERSION TO PRECALCINER KILN	4.50	LB/T CLINKER	3.60	LB/T	4.50
IN-0081	LONE STAR INDUSTRIES, INC.	IN	133-10159	09/18/1998 ACT	04/16/1999 ACT	CEMENT MANUFACTURING, KILN OPERATION	LOW NOX CALCINER, GOOD COMBUSTION PRACTICES	4,428.00	T/YR	5.47	LB/T	5.47
IN-0112	LONE STAR INDUSTRIES, INC.	IN	133-5886-00002-3241	05/16/1996 ACT	09/18/1998 ACT	CEMENT KILN, WET PROCESS, COAL	LOW NOX BURNERS AND GOOD COMBUSTION	471.00	LB/H	-	-	6.28
MD-0027	LEHIGH PORTLAND CEMENT COMPANY	MD	06-6-0356R	02/05/1997 ACT	06/08/2000 ACT	CEMENT MANUFACTURING, PREHEATER/PRECALCINER KILN	A 5-STAGE PREHEATER/PRECALCINER PYROPROCESSING PLANT IS SELECTED FOR NOX EMISSION ABATEMENT. ANY ADD-ON NOX EMISSIONS CONTROL HAS BEEN DETERMINED TO BE EITHER TECHNICALLY OR ENVI. INFEASIBLE	4,871.00	T/YR	-	-	4.40
MI-0287	HOLNAM, INC.	MI	60-71L	01/11/2000 ACT	03/20/2000 ACT	CEMENT KILNS, WET PROCESS (2)	CURRENT EXISTING LIMIT DOES NOT REFLECT POTENTIAL 30% REMOVAL IN SLURRY-SCRUBBER. SOME GENERATION OF NO2 BY NATURAL GAS COMBUSTION IN RTOS.	6.00	LB/T	-	-	6.00
MI-0354	HOLNAM, INC	MI	60-71K	09/15/1997 ACT	06/23/1998 ACT	CEMENT KILNS, WET PROCESS, (2)	-	3,377.00	T/YR	6.00	LB/T	6.00
TN-0086	SIGNAL MOUNTAIN CEMENT COMPANY, LP	TN	47-065-3070	01/09/1998 ACT	05/29/1998 ACT	MANUFACTURING, PORTLAND CEMENT, DRY FEED KILN	GOOD COMBUSTION PRACTICES	1,500.00	PPM	403.00	LB/H	2.52
TX-0279	NORTH TEXAS CEMENT COMPANY	TX	PSD-TX-893	12/05/1997 ACT	03/04/1999 ACT	MAIN KILN/SCRUBBER STACK	SEPARATE LINE DOWNDRAFT LOW-NOX CALCINER AND LOW-NOX IN-LINE CALCINER.	1,085.00	LB/H	1,585.00	T/YR	8.40
TX-0282	CAPITOL CEMENT DIVISION	TX	PSD-TX-120M3	06/24/1997 ACT	09/16/1998 ACT	DRY KILN EXHAUST BAGHOUSE (KS-1A)	NONE INDICATED.	450.00	LB/H	1,971.00	T/YR	5.40
TX-0282	CAPITOL CEMENT DIVISION	TX	PSD-TX-120M3	06/24/1997 ACT	09/16/1998 ACT	ALKALI BYPASS BAGHOUSE STACK (9A)	NONE INDICATED.	50.00	LB/H	219.00	T/YR	Cannot be determined from RBLC data.
TX-0282	CAPITOL CEMENT DIVISION	TX	PSD-TX-120M3	06/24/1997 ACT	09/16/1998 ACT	WET KILN EXHAUST BAGHOUSE (KS-1B)	NONE INDICATED	450.00	LB/H	1,971.00	T/YR	10.41
TX-0282	CAPITOL CEMENT DIVISION	TX	PSD-TX-120M3	06/24/1997 ACT	09/16/1998 ACT	DRY/WET KILN & ALKALI BYPASS BAGHOUSE STACK (KS-1)	NONE INDICATED	950.00	LB/H	4,161.00	T/YR	Cannot be determined from RBLC data.

¹ Per search performed on May 2, 2011. The following is a list of search criteria inputs.

(1) Permit Dates: January 1, 1997 to December 31, 2000.
(2) Process Type 90.028 - Portland Cement Manufacturing and Process Name Contains: Kiln
(a) Process Type: 90.028 - Portland Cement Manufacturing
(b) Process Name Contains: Kiln
(3) Pollutant Name: NO_x
(4) Facility State: All States

² Missing or 0 data has been replaced with "-".

³ A throughput value is available; however, units are not specified. Therefore, NO_x lb/ton clinker emission rate cannot be determined from RBLC data.

⁴ Emission limit is based on lb/ton kiln feed rate. Clinker data is unavailable; therefore, NO_x lb/ton clinker emission rate cannot be determined from RBLC data.

APPENDIX A. QUALIFICATIONS SUMMARY

HOFMANN, JOHN
President & CEO
Trinity Consultants

EDUCATION

B.S. Chemical Engineering, MIT
M.S. Chemical Engineering, MIT
M.S. Atmospheric Science, the University of Arizona

SUMMARY

With my experience, I have taught numerous professional courses on various air quality regulatory subject matters relative to Federal Clean Air Act (CAA) requirements, PSD, and state equivalent programs. Attendees of my courses have included both the regulated source community and also state and U.S. EPA representatives.

I serve as the President and Chief Executive Officer of Trinity Consultants, Inc., at the firm's head quarters office in Dallas, Texas. As a 25-year veteran of the firm, I have served in several key roles at Trinity, a 300-person environmental consulting and software firm with 30 offices world-wide. The firm is strongly focused on industrial CAA compliance work, with a heavy emphasis involving permitting of new sources including sources at cement plants. Before being selected as President and CEO, I served as the firm's COO.

I have substantial consulting experience and am considered as an authority on CAA regulations, air pollution (source) impact assessments, air pollution control technologies, and environmental management systems. I have also conducted process development work associated with air pollution controls used in oil refining operations and conducted research of the formation of acid rain. My consulting focus is CAA compliance. I have routinely been called upon to formally evaluate prospective capital projects involving upgrades and implementation of new and more efficient process in a variety of ways. I have also created and taught numerous private and customized CAA classes to major industrial companies.

My compensation rate in preparing this report was \$285 per hour. I never before have not testified made any testimony as an expert at a trial or by deposition.

APPENDIX B. DOCUMENTS REVIEWED

REGULATIONS

1. 40 C.F.R. 52.21 (current)
2. 40 C.F.R. 52.21 (Dec. 31, 2002)
3. 40 C.F.R. 52.21 (July 21, 1992)
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5. 45 Fed. Reg. 52,676 (Aug. 7, 1980)
6. 57 Fed. Reg. 32,314 (July 21, 1992)
7. 67 Fed. Reg. 80,186 (Dec. 31, 2002)
8. 71 Fed. Reg. 54,235 (Sept. 14, 2006)
9. 5 COLO. CODE REGS. §1001-2 (Common Provisions Reg.) (1995)
10. 5 COLO. CODE REGS. §1001-5 (Reg. No. 3) (1995)
11. 5 COLO. CODE REGS. §1001-5 (Reg. No. 3), Part A (1998)
12. 5 COLO. CODE REGS. §1001-5 (Reg. No. 3), Part B (1998)
13. *Wisconsin Electric Power Co. v. Reilly*, 893 F.2d 901 (7th Cir. 1990)

DOCUMENTS

14. Applicability of New Source Review Circumvention Guidance to 3M-Maplewood, Minnesota (U.S. EPA, June 17, 1993)
15. Accomplishing 2000 Production Plan, CMXLYO00157265-82
16. Air Pollutant Emission Notices, 1993
17. Air Pollutant Emission Notices, 1994, CEMEX_NEIC_00002732
18. Alternative Control Techniques Document Update - NOx Emissions from New Cement Kilns, U.S. Environmental Protection Agency, November 2007 (EPA-453/R-07-006)
19. Amended Answer of Defendant CEMEX, Inc., to Plaintiff's First Amended Complaint, May 6, 2009, Document 96

20. AP-42 Section 11.6, Table 11.6-8, January 1995
21. Authorization for Expenditure, February 1998, CEMEX_NEIC_00003321-34
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25. CDPHE Construction Permit Belt No. 94-BO-593, September 11, 1998, CEMEX_NEIC_00000208-14
26. CDPHE Construction Permit No. 12BO444-1, December 17, 1993, US00006530-33
27. CDPHE Construction Permit No. 12BO444-2, December 17, 1993, CEMEX_NEIC_00000203-07
28. CDPHE Construction Permit No. 93BO1414F, March 15, 1995, CMXLYO00607471-75
29. CDPHE Construction Permit No. 94BO593, March 15, 1993, CMXLYO00607476-78
30. CDPHE Construction Permit No. 98-BO-0259, September 11, 1998, CEMEX_NEIC_00000246-56
31. CDPHE Construction Permit No. 98-BO-0315, September 11, 1998, CEMEX_NEIC_00000277-84
32. CDPHE Construction Permit No. 98-BO-1414-F, September 11, 1998, CEMEX_NEIC_00000257-62
33. CDPHE Handwritten notes, January 6, 1998, US00006344-45
34. CDPHE Lyons Cement Plant Finish Cement Grinding System Application Process History, 1998, US00006500-01
35. CDPHE Operating Permit, Title V Permit, February 1, 2000, US00003427-538
36. CDPHE Organizational Chart, 1998, CMXLYO00038860
37. CDPHE Preliminary NSR Analysis, July 13, 1998, US00006201-02
38. CDPHE, Preliminary Analysis Permit No. 12BO444-2, March 6, 1989, CMXLYO00425024
39. CEMEX Construction Permit and Permit Modification Application, April 13, 1998, CMXLYO00015037-61

40. CEMEX Lyons Cement Plant Presentation, CMXLYO00150942-70
41. CEMEX Lyons Facility, Estimated Clinker Production - Historical Summary, CMXLYO00628849-852
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43. CEMEX Lyons Facility, Oxygen Injection - Historical Summary, CMXLYO00628853-855
44. CEMEX Lyons Historic NOx Emission Trends, 2001-2007, CMXLYO00131241-50
45. CEMEX, Inc., Lyons Cement Plant, Capital Projects 1992 – 1999 (Redacted) CMXLYO00180114-116
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50. David Sanders, EPA, NOx Control Technologies for the Cement Industry (Final Report), September 19, 2000
51. Defendant CEMEX, Inc., Responses to United States' First Set of Interrogatories, September 4, 2009
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53. Downtimes, 2001-2006, CEMEX_NEIC_00003872-5586
54. Drawing, September 19, 2000, CMXLYO00402670
55. Email from John Lohr to Dave Repasz, Re: power review.xls, September 19, 2000, CMXLYO00045547-48
56. Email From John Lohr to Steve Bryan, Re: Lyons Kiln Outage, March 3, 1999, CMXLYO00045705
57. Email from John Lohr to Steve Bryan, Re: Lyons Kiln Status, August 17, 2000, CMXLYO00157315
58. Email from McCabe to Schnarre re Clinker, October 17, 2000, CMXLYO00402668-69
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66. Facsimile from Keith Baugues to John Lohr, Re: Please see attached, March 11, 1998, US00000080-82, 90
67. Fan Performance Curve, CMXLYO00628857
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72. Fan Specifications and Performance Curve, CMXLYO00628883
73. Fan Specifications, CMXLYO00628865
74. Fan Specifications, CMXLYO00628865
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79. H. Tseng, P. Alsop “Apparatus for Oxygen Enrichment of Cement Kiln System,” U.S. patent 6,688,883, February 10, 2004 (and associated files)
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97. Letter from Amarjit Gill to Dennis Meyers, CDPHE, Re: Kiln Permit No. 12BO444-2 Compliance Demonstration Methodology, March 20, 1998, US00027808-12
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100. Letter from Amarjit Gill to James King, Re: Draft Operating Permit for Southdown, Inc.'s Lyons Cement Plant, August 25, 1999, CMXLYO000120210-13
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109. Letter from Keith Baugues to John Lohr, Re: Kiln and Dryer Emission Estimates: MMA Project Number 786-98, March 12, 1998, CMXLYO00038872-75
110. Letter from Keith Baugues to John Lohr, Re: Update on Calculating Emissions, March 10, 1998, US00062232-34
111. Letter from Larry Byrd, BOC Gases, to Andrew Jackura, Re: Request for Quotation, January 9, 1998, CEMEX_NEIC_00003626-654
112. Letter from Ram Seetharam to Amarjit Gill, Re: Draft Permits, June 12, 1998, CMXLYO00015080-118
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127. Memorandum from Dennis Myers, Re: The ghost of Holnam, January 6, 1998, US00006499
128. Memorandum from Eric Leigh to John Lohr, Re: Reserve numbers, March 6, 1998, US00035105-06
129. Memorandum from Gregg St. Clair to Dave Repasz, Re: Lyons Plant Expansion – Phase I, November 19, 1997, CEMEX_NEIC_00003291-96
130. Memorandum from Gregg St. Clair to Dave Repasz, Re: Plant Expansion Project Update, March 26, 1998, CMXLYO00155748-82
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APPENDIX C. RECENT PUBLICATIONS

Recent Publications for John Hofmann, P.E. In Last 10 Years

Conference Presentations and Corporate/Governmental Entity Workshops

1. "NSR Applicability and Clean Air Act Enforcement for Electric Utilities" - Entergy Arkansas, Little Rock, AR, April 11, 2000
2. "Prevention of Significant Deterioration (PSD) Short Course" - Navistar International Transportation, Chicago, IL, May 17-18, 2000
3. "Clean Air Act Workshop" - US EPA Office of Atmospheric Programs - Washington, DC, July 24-25, 2000
4. "Clean Air Act - Title V Permits" - Trigen Energy - Trigen EHS Conference 2000, Baltimore, MD, October 6, 2000
5. "Clean Air Act Workshop" - Tampa Electric Company, Tampa, FL, October 9-10, 2001
6. "Prevention of Significant Deterioration (PSD) Short Course" - Honda of America Manufacturing, Inc. - Columbus, OH, June 18-19, 2002
7. "New Source Review Reform" - Chemical Lime Annual Environmental Meeting- Fort Worth, TX, January 22, 2003
8. Enforcement Session Moderator - AWMA NSR Reform Conference - Crystal City, VA, April 29, 2003
9. "Strategic Management of the NSR Permitting Process" - IDEA 18th Annual Campus Energy Conference, 2005
10. "US Air Quality Regulations" - Engelhard Corporation, Iselin, NJ, June 9, 2005
11. "MACT Compliance - Nissan North America Executive Summary" - Smyrna, TN, July 26, 2005
12. "Environmental Risk Assessment - Methods and Analysis" - 2005 CIBO Annual Meeting - Newport, RI, October 14, 2005
13. "Air Quality Regulations & Permitting" - Ford Kansas City Assembly Plant - Claycomo, MO, January 25-26, 2006
14. "Air Quality and Pollution Control" - ABA Environmental Sciences Program (Webinar), February 9, 2006

15. "NSR/NSR Reform Workshop" - Kinder Morgan Limited Partners - Orange, CA, April 17-19, 2006
16. "Environmental Session - Council of Industrial Boiler Owners (CIBO) Issues Briefing and Discussion" - CIBO Annual Meeting - San Diego, CA, October 27, 2006
17. "NSR/NSR Reform Workshop" - US STEEL Corporation - Birmingham, AL, February 26-27, 2007
18. "The Evolution of BACT" - Environmental Federation of Oklahoma, Oklahoma City, OK, October 12, 2007
19. "GHG Developments and Practical Strategies for Addressing Climate Change" - CIBO Annual Meeting - Phoenix, AZ, October 19, 2007
20. "PSD Workshop" -Tampa Electric Company, Tampa, FL, November 30, 2007
21. "Workshop on Green House Gas (GHG) Regulatory Developments, NSR/GHG Interaction," USEPA Office of Air Quality Planning and Standards, RTP, NC, January, 9, 2008
22. "Emission Reduction – Sustainable Development for Process Industries" - Environmental Workshop, Dubai, UAE, February 8-9, 2009
23. "NSR/PSD Workshop" - Covidien Corporation - St. Louis, MO, March 12-13, 2009
24. "Advanced NSR - Western States Air Resources (WESTAR) Council NSR Workshop" - Las Vegas, NV, Boise, ID, and Lacey, WA, (3 separate workshops in 2010)
25. "Final PSD/Title V Tailoring Rule - Applicability and Ramifications" - Dallas, TX, June 10, 2010
26. "Transitional PSD Applicability Scenarios"- Troutman Sanders - PSD Meeting/Update - Atlanta, GA, July 20, 2010
27. "An Update on Recent Changes to the Texas SIP" - Gas Processors Association of Houston - Regional Meeting - Houston, TX October 12, 2010
28. "Advanced NSR/PSD Short Course" - Michigan Department of Environmental Quality - Lansing, MI, May 23-25, 2011

Papers and Publications

1. Schulze, Richard and Hofmann, Jay, PSD Applicability: Routine Maintenance & Emissions Change Issues, White Paper, February 9, 2006.
2. Hofmann, John, E., "U.S. Industrial Air Pollution Regulations, Controls, Surveillance, and Assessments," Environmental Sciences: A Primer for the Technically Challenged (and an Update for the Technically Sophisticated), American Bar Association Section of Environment, Energy, and Resources, November 3, 2004.

3. Hofmann, John, E. (Reviewer), "New Source Review For Stationary Sources of Air Pollution," National Research Council of the National Academies, March 8, 2006

APPENDIX D. SUPPORTING CALCULATIONS

Evaluation of Change in NOX Emissions as a Result of Oxygen Enrichment**Question:** Does the use of Oxygen Enrichment increase NOX emissions?**Null hypothesis:** The average NOX emissions with no Oxygen Enrichment is the same as with Oxygen Enrichment**Test:** Student's t Test for Independent Samples (Unpaired)
$$t = \frac{\bar{X}_1 - \bar{X}_2}{s_{\bar{X}_1 - \bar{X}_2}}$$
One-tailed test, $p < 0.05$, unequal sample sizes, unequal variancesSatterthwaite Approximation
$$s_{\bar{X}_1 - \bar{X}_2} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$
Degrees of Freedom
$$d.f. = \frac{(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2})^2}{(\frac{s_1^2}{n_1})^2/(n_1 - 1) + (\frac{s_2^2}{n_2})^2/(n_2 - 1)}$$

		1 CEMS NOX (lb/hr) No OE	2 CEMS NOX (lb/hr) With OE	1 CEMS NOX (lb/ton of feed) No OE	2 CEMS NOX (lb/ton of feed) With OE
Arithmetic Mean	E	555.0684	472.9072	6.2949	4.7366
Maximum		687	643	8	7
Minimum		462	274	4	3
Sample Size	n	117	334	117	334
Sample Variance	S ²	2549.2022	3368.5109	0.5954	0.5143
Sample Std. Deviations	S	50.4896	58.0389	0.7716	0.7172
Satterthwaite Approx.	SE _s	5.65		0.08	
Test Statistic	t	14.55		19.14	
Degrees of freedom	df	231		191	
Critical Value of t (for 95% confidence)	t'	1.651		1.653	
Is E2 > E1		no		no	
Is t > t'		yes		yes	

Null Hypothesis	Population Mean of 1 = Population Mean of 2		
Reject Null Hypothesis	If $t > t'$	Reject	Reject
Statistically Significant		Decrease	Decrease

Assumptions: Population is approximately normally distributed

Used Studentized Range Test to support the assumption that the population is normally distributed by evaluating if the sample sets are normally distributed.

$$\text{Compare } \frac{w}{s} = \frac{X_{(n)} - X_{(1)}}{s} \text{ to critical values}$$

Range	w	225	369	3	5
Sample Std. Deviations	S	50.4896	58.0389	0.7716	0.7172
Studentized Range Test		4.4564	6.3578	4.5218	6.8800
Upper Critical Value	$\alpha = 0.01$	6.4552	7.0991	6.4552	7.0991
Lower Critical Value	$\alpha = 0.01$	4.1952	4.8312	4.1952	4.8312
Normal?		normal	normal	normal	normal

Summary of Tests of Hypotheses for the Regression Analysis of Single Variables

For pure fuel, model used:

$$Y = b_0 + b_1 X_1 + b_2 X_2 + e.$$

Null/ Alternative Hypothesis for Global Usefulness

$$H_0: b_0 = 0, b_1 = 0, b_2 = 0$$

 $H_1: \text{One of } b_i\text{'s is not } 0$

Null/ Alternative Hypothesis for Individual regression coefficient

$$H_0: b_k = 0$$

 $H_1: b_k \text{ is not } 0$

Data	Independent Variables	Dependent Variables	Probability for the Test of Null Hypothesis for Global Usefulness	Parameters	Parameter Values	Probability for the Test of Null Hypothesis for Individual regression coefficient
No OE	Coal	NOX	0.000693	b_0	-127.143701	0.517193
				b_1	110.230232	0.000693
No OE	O2	NOX	0.000810	b_0	412.267833	0.000000
				b_1	61.932993	0.000810
No OE	Feed	NOX	0.000055	b_0	857.465337	0.000000
				b_1	-3.410887	0.000055
OE	Coal	NOX	0.000001	b_0	729.337609	0.000000
				b_1	-35.508360	0.000001
OE	O2	NOX	0.024385	b_0	432.715670	0.000000
				b_1	18.760347	0.024385
OE	Feed	NOX	0.000001	b_0	733.960082	0.000000
				b_1	-2.600551	0.000001
All	Coal	NOX	0.000000	b_0	908.511903	0.000000
				b_1	-59.577689	0.000000

Data	Independent Variables	Dependent Variables	Probability for the Test of Null Hypothesis for Global Usefulness	Parameters	Parameter Values	Probability for the Test of Null Hypothesis for Individual regression coefficient
All	O2	NOX	0.000001	b ₀	399.223965	0.000000
				b ₁	43.482312	0.000001
All	Feed	NOX	0.000000	b ₀	949.722510	0.000000
				b ₁	-4.679417	0.000000
All	OE	NOX	0.000000	b ₀	526.178042	0.000000
				b ₁	-0.001029	0.000000